Following the Stars: Long-Run Rates in the Carry Trade*

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Abstract

Persistent cross-country differences in long-run interest rates command a currency risk premium beyond the carry trade. Motivated by the Fisher equation, we estimate each country's long-run nominal rate in real time as the fair value of its 10-year yield given inflation and growth, and sort currencies on its differential to build *carry**. *Carry** earns 3.52% a year, with a higher Sharpe ratio than traditional and static carry, a significant alpha over both and over other currency factors, and a significant price of risk in the cross-section. Unlike carry, whose return is mostly static, two thirds of its return comes from updating positions as long-run rates move, and exchange rates do not offset the long-run part of interest differentials. The premium comes from the inflation part of the long-run rate, while the real part is not priced. *Carry** loses less than carry in bad times but is more exposed to option-implied crash risk and oil risk.

Keywords: Carry trade; currencies; long-run interest rates; long-run inflation; Fisher equation.

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1. Introduction

The carry trade anomaly is one of the key stylized facts in international finance, reflecting deviations from uncovered interest parity. It refers to a profitable trading strategy that involves borrowing in low interest-rate currencies while lending in high interest-rate currencies, documented at least since the 1980s (Hansen and Hodrick, 1980; Fama, 1984). The profitability of this strategy arises because the positive returns due to interest rate differentials are only partially offset by currency depreciation. The benchmark empirical representation of this strategy is the carry factor of Lustig et al. (2011), which sorts currencies on nominal short-term interest rates.

Recent work by Hassan and Mano (2019) concludes that sorting countries by interest rate differentials can be well approximated by a static ranking, implying that a country with higher interest rates tends to maintain this position relative to a lower-rate country, even though both may experience rate fluctuations over time. Therefore, understanding long-term interest rate differences across countries is crucial for explaining carry trade profits. We refer to their static ranking strategy as static carry.

Building on this insight, we sort currencies on the differences in their long-run nominal interest rates. Sorting on long-run differentials but implementing in short-term forward contracts represents a middle ground between the traditional carry factor of Lustig et al. (2011) and the static ranking interpretation of Hassan and Mano (2019). Unlike traditional carry, our signal isolates the long-run component of interest rate differentials. At the same time, it is more flexible than a purely static ranking, since it allows countries to change positions over time as their long-run rates evolve, while still filtering out short-term movements. We refer to the resulting factor as $carry^*$, since it is based on the long-run nominal interest rate typically denoted i^*.

By the Fisher equation, the long-run nominal rate of a country is the sum of its long-run expected inflation and its long-run real rate, $i^* = \pi^* + r^*$. Neither component is observed in real time across countries. We therefore estimate i^* as the fair value of the 10-year government bond yield given inflation and output growth. Each month, we regress the 10-year yields of all countries on their past 1- and 3-year inflation and 1- and 5-year GDP growth, using only the data vintages available at the time, and take the fitted value as i^*. Inflation proxies the long-run expected inflation and growth proxies for the long-run real rate, which Laubach and Williams (2003) link to trend growth. A similar estimation is used by Poghosyan (2014), who relates long-run sovereign bond yields in advanced economies to potential growth and public debt. Our estimate differs in that it is constructed in real time, and we use it to rank currencies rather than to assess bond

yields.

Our contribution is fourfold. First, we show that $carry^*$ is a distinct signal that earns a large premium. It is far more persistent than carry, with a first-order autocorrelation of 0.91 against 0.53, and its rank correlation with the forward premium is 0.55. As a result, $carry^*$ turns over 4.82 times a year, against 10.89 for carry. We further show that the estimated long-run rates are closely related to the two main drivers of natural interest rates, productivity and demographics (Carvalho et al., 2016; Cesa-Bianchi et al., 2022; Ferreira and Shousha, 2023; Obstfeld, 2023). Since the fair value uses GDP growth, part of the link with productivity is mechanical, but it also holds for π^*, which uses inflation only. A simple strategy which goes long currencies with above-median long-run differentials and shorts those below delivers a statistically and economically significant annualized excess return of 3.52%. Its Sharpe ratio is 0.57 before and 0.54 after quoted bid-ask spreads, against 0.35 and 0.29 for carry and 0.30 for static carry. Relative to both benchmarks, $carry^*$ also has higher Sortino and Omega ratios and less excess kurtosis, and its skewness is less negative than that of carry.

Second, we show that $carry^*$ is a priced factor that subsumes carry and static carry. Fama-MacBeth regressions on two sets of G10 test assets, our own 27 portfolios sorted on the three signals and the 45 currency portfolios of Della Corte and Riddiough (2025), give $carry^*$ an annualized risk price of 2.74% and 3.47%. The estimates are 3.02% and 3.73% when carry and static carry enter the same regression, while neither benchmark commands a significant risk premium once $carry^*$ is included. Time-series spanning regressions reveal an asymmetric relationship, in which $carry^*$ earns a significant abnormal return of 1.93% per year after controlling for carry, static carry and the dollar factor, while the abnormal returns of both benchmarks are economically small and statistically insignificant once $carry^*$ is included. The abnormal return of $carry^*$ also survives currency value and momentum, the term spread, global imbalances, country size and commodity factors.

Third, we show that the inflation part of the long-run rate is what is priced. Splitting the 10-year yield along the Fisher equation, currencies sorted on π^* earn Sharpe ratios of 0.53 to 0.54 and a significant abnormal return over carry and static carry, and π^* carries a significant risk price next to carry on both sets of test assets. Currencies sorted on the real part earn Sharpe ratios of 0.23 to 0.25, no abnormal return and no significant risk price, even though the real part accounts for 70% of the cross-sectional variance of nominal 10-year differentials in the period covered by inflation swaps. This also explains why $carry^*$ improves on the 10-year yield itself, whose sort earns a Sharpe ratio of 0.39 and no abnormal return over carry and static carry. The evidence is consistent with compensation

for long-run inflation risk.

Fourth, we show where the return of $carry^*$ comes from and what risks it carries. $Carry^*$ earns a smaller interest differential than carry, 2.49% against 4.09% a year, but gains 0.98% a year through currency movement, where carry loses 1.96%. In uncovered interest parity regressions, the exchange rate offsets most of the forward premium that is unrelated to the long-run rate, with a slope of 0.88, but not its long-run part, with a slope of -0.07. The risk of $carry^*$ has two dimensions. It is smaller in bad times, as $carry^*$ earns 1.51% a year in US recessions, where carry loses 5.86%, and loses 23.14% in the worst 10% of months for carry, against 39.95% for carry itself. However, $carry^*$ is not less exposed to option-implied crash risk and oil risk, as its betas to changes in risk reversals and in oil implied volatility are larger than those of carry.

Our paper contributes to the literature seeking to explain the sources of carry trade profitability. Brunnermeier et al. (2008) link carry profits to exposure to global liquidity risk, while Menkhoff et al. (2012a) emphasize the role of innovations in global FX volatility. Ready et al. (2017) argue that carry returns reflect the structural dependence of commodity-importing versus commodity-exporting countries, and Della Corte et al. (2016) relate currency premia to global imbalances. Other work documents currency momentum (Menkhoff et al., 2012b) and currency value (Asness et al., 2013; Menkhoff et al., 2017). Our contribution is complementary: we show that the profitability of carry trades is associated with persistent cross-country differences in long-run interest rates, and in particular with their inflation part.

Two studies also use macroeconomic or yield curve information beyond the short rate to predict currency returns. Dahlquist and Hasseltoft (2020) show that past trends in economic activity and inflation predict currency excess returns. Their signal is the recent change in macroeconomic fundamentals, whereas we sort on the level of the long-run interest rate that fundamentals imply. Ang and Chen (2010) show that the shape of the yield curve contains information about future currency returns. A factor sorted on the term spread earns -0.30% a year in the data, and $carry^*$ keeps an abnormal return of 3.26% after controlling for it, which separates the long-run rate from the slope of the yield curve.

More closely related to our study is Lustig et al. (2019), who examine currency portfolios formed by sorting on the excess returns of long-term bonds while also implementing the carry trade strategy in long-term bonds. They find that the profitability of currency carry trades declines with the maturity of the underlying foreign bonds. Our approach differs in at least three important aspects. First, rather than sorting on excess bond returns, we rank

currencies based on long-run interest rate differentials. Second, while their analysis focuses on long-term bonds, we implement the carry trade using short-term forward contracts. Third, unlike Lustig et al. (2019), we split the long-run rate into its inflation and real parts and show that only the inflation part is priced.

Our results also complement Feunou et al. (2024), who show that, under no arbitrage, persistent differences in interest rate trends across countries must be matched by a persistent currency risk premium, and who find that carry trades earn more when the average trend differential to the US is large. While they estimate trends from a two-country term structure model and study the time variation of carry returns, we sort currencies on real-time long-run rates in the cross-section, show that the resulting factor is priced and subsumes carry, and locate its premium in the inflation part of the long-run rate. Moreover, unlike the return of carry, which Hassan and Mano (2019) and Feunou et al. (2024) find to be mostly static, two thirds of the return of $carry^*$ comes from changing positions over time as long-run rates move.

This paper also relates to the literature on the asset-pricing implications of long-run trends in nominal interest rates. Kozicki and Tinsley (2001) and Cieslak and Povala (2015) show that persistent inflation trends account for important variation in US treasury yields, while Bauer and Rudebusch (2020) combine inflation and real-rate trends to construct a nominal trend component that subsumes earlier measures and provides a powerful explanation of yield dynamics. Instead of focusing on US treasury yields, we use long-run trends in nominal interest rate differentials to explore FX implications.

The paper is structured as follows. Section 2 presents our data, defines the long-run rate, describes the portfolio formation and the benchmark carry strategies and establishes a common sample across all strategies. Section 3 studies $carry^*$ in the cross-section of currency returns. It establishes that $carry^*$ is distinct from carry, documents its excess return and its performance relative to the benchmarks, and assesses its pricing ability through Fama-MacBeth and spanning regressions. It then examines the stability of the results across subperiods and states, and compares $carry^*$ with other currency factors. Section 4 studies where the premium comes from, splitting the return of $carry^*$ into the interest differential and the currency movement and the long-run rate into its inflation and real parts, and examines the risk of $carry^*$ in bad times and its exposure to crash risk and oil risk. Section 5 discusses the robustness of our results, and Section 6 concludes.

2. Data, long-run rate and portfolio formation

This section describes the currency, interest rate and macroeconomic data used in the empirical analysis, the estimation of the long-run rate, the construction of portfolios and associated excess returns, and defines the benchmark carry measures for comparison together with the common sample.

2.1. Data

We obtain data for spot exchange rates and 1-month forward exchange rates versus the US Dollar (USD), spanning from October 1983 to February 2025, from Datastream. Similar to Lustig et al. (2019), the empirical analysis focuses on G10 countries. Namely, we focus on the United States (US), Australia (AU), Canada (CA), Great Britain (GB), Japan (JP), Norway (NO), New Zealand (NZ), Sweden (SE), Switzerland (CH), Germany (DE) and the Euro Area (EU). We keep the Deutsche mark prior to January 1999 and the euro thereafter, consistent with standard practice in the FX literature. The FX data for Australia and New Zealand begin in December 1984.

Net returns use quoted bid and ask prices from Datastream. For each currency and month, the half-spread is the monthly median of the daily values of $(ask - bid)/(ask + bid)$. Quotes for the 1-month forward rate begin in May 1990. For the Deutsche mark, no forward quote exists, and we construct its forward half-spread as its spot half-spread, available from October 1989, plus the gap between the forward and the spot half-spreads of the synthetic euro. Before the first quote, each currency takes the median of its first 36 quoted months.[1]

The 10-year benchmark government bond yields are from Datastream, extended back with OECD long-term interest rates where the Datastream series start later. The yields of the US, Canada, Switzerland, Germany and Japan are available from January 1970, which gives a long history for the estimation of the long-run rate described in Subsection 2.2. For the Euro area, the 10-year benchmark yield prior to January 1999 is sourced from a Datastream synthetic composite series, which aggregates member country government bond yields. Sweden's 10-year yield series begins in January 1987 and cannot be extended, as earlier data are unavailable from the OECD.

The macroeconomic data are consumer prices (CPI) and real GDP from the OECD. To avoid look-ahead bias, we use the data as they were available at each point in time. At the

[1]Quoted spreads are generally wider than the spreads at which dealers trade, so our net returns are conservative. For Canada and Switzerland, the spot half-spread before November 2000 is set to the forward half-spread. Appendix A reports the half-spreads by currency and decade, and the turnover and trading cost of each strategy. Trading costs are 0.18% per year for $carry^*$, against 0.39% for carry.

end of each month t from February 1999, we take the data from the edition of the OECD Main Economic Indicators Original Release and Revisions Database published in month t. No vintages are available before 1999, and for those months we use the latest data lagged by their publication delay, one month for CPI and three months for GDP. From July 2022, the Japanese CPI in the OECD vintages is no longer updated, and we use the national CPI from Datastream with a two-month lag, the typical age of the Japanese data in the vintages. From these data we compute, for each country and month, annualized CPI inflation over the past one and three years and annualized real GDP growth over the past one and five years, all as log changes in percent. These variables start in January 1970 for all countries except the Euro area, computed from OECD series that begin in 1955 or earlier for CPI and around 1960 for GDP. The Euro area series begin in 1990 for CPI and 1991 for GDP, so its variables are available from June 1996, and its vintages from October 1999.

Section 3 uses further data. The test assets of the Fama-MacBeth regressions include the 45 G10 currency portfolios of Della Corte and Riddiough (2025). Breakeven inflation rates, 25-delta risk reversals and implied volatilities of currency and oil options are from LSEG, and the NBER recession indicator and the VIX are from FRED. Each of these is described where it is used.

2.2. The long-run rate

By the Fisher equation, the long-run nominal rate of a country is the sum of its long-run expected inflation and its long-run real rate, $i^* = \pi^* + r^*$ (Bauer and Rudebusch, 2020). We estimate i^* as the fair value of the 10-year yield given inflation and output growth. At the end of each month t, we estimate by ordinary least squares (OLS) the pooled regression

$$i_{c,s} = \beta_0 + \beta_1 \pi_{c,s}^{(1)} + \beta_2 \pi_{c,s}^{(3)} + \beta_3 g_{c,s}^{(1)} + \beta_4 g_{c,s}^{(5)} + \varepsilon_{c,s}, \tag{1}$$

where $i_{c,s}$ is the 10-year yield of country c in month s, $\pi_{c,s}^{(k)}$ is CPI inflation over the past k years and $g_{c,s}^{(k)}$ is real GDP growth over the past k years. The regression pools all countries, including the US, and all months $s \leq t$, so the estimation window expands over time. The first estimate requires at least 60 months of pooled data, a condition met before our sample starts, since the pooled data begins in 1970. A country therefore has a long-run rate estimate as soon as it enters the panel. Inflation proxies π^*, and growth proxies r^*, which Laubach and Williams (2003) link to trend growth. We measure the long-run rate $i_{c,t}^*$ as the fitted value estimated with data up to month t. Crucially, both the coefficients and the macroeconomic data use only information available at the end of month t, so $i_{c,t}^*$

is a real-time quantity free of look-ahead bias. The regression has no country fixed effects and no yields on the right-hand side, so the differences in i^* across countries come only from differences in inflation and growth.[2] Figure 1 plots the cross-section of $i^*_{c,t}$ over the sample, which can be seen to have a downward trend over the past decades, similar to what is documented by Bauer and Rudebusch (2020), with a reversion after 2020.[3]

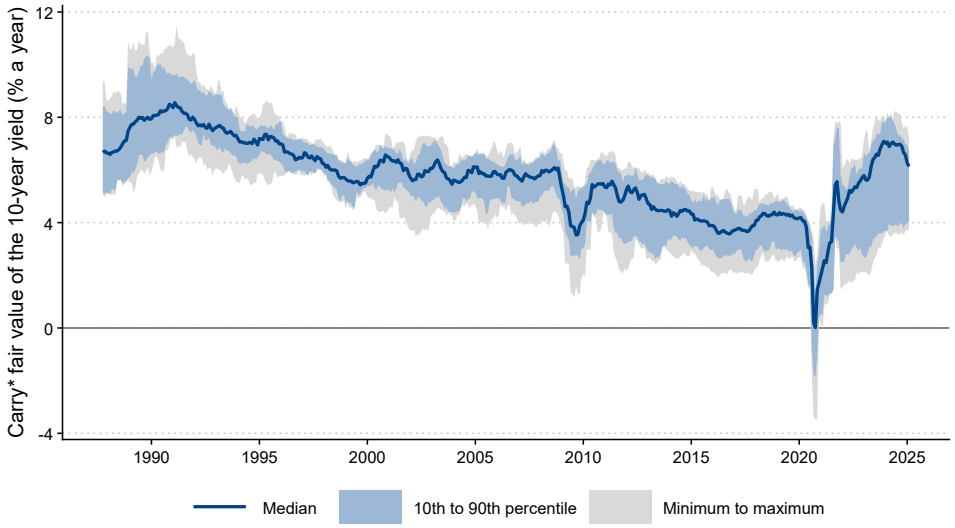


Figure 1: Cross-section of the $carry^*$ fair value. Each month, the line plots the median across the US and the nine currencies of the $carry^*$ fair value of the 10-year yield, $i^*_{c,t}$, in levels. The dark band covers the 10th to 90th percentiles, and the light band the minimum to the maximum. Sample October 1987 to February 2025, in % a year.

2.3. Portfolio formation

We use the long-run rate of Subsection 2.2 to compute our signal, the long-run differential in nominal interest rates with respect to the US,

$$carry^*_{c,t} = i^*_{c,t} - i^*_{US,t}. \tag{2}$$

At the end of each month t, we allocate currencies to two portfolios based on their long-run differential with respect to the US at the end of month t. Currencies are ranked from low to high values of our sorting variable, and the split is made at the median. Therefore, portfolio 1 (P1) contains currencies with the lowest values of $carry^*_c$ and portfolio 2 (P2) contains currencies with the highest values of $carry^*_c$. We compute the (log-)excess currency

[2] Appendix A defines 24 alternative measures of the long-run rate, which change the macro features, the horizons, the target of the regression and the estimator, and Appendix C shows that the main results hold for each of them.

[3] Appendix I shows that $i^*_{c,t}$ is also related to the main drivers of natural interest rates, as it moves with demographics and productivity growth within each country.

return, $rx_{c,t+1}$, as the value of buying a foreign currency in the (log-)forward market, $f_{c,t}$, and then selling it in the (log-)spot market after one month, $s_{c,t+1}$. Since covered interest parity holds closely in the data at the monthly frequency as shown in Akram et al. (2008), our excess return corresponds to

$$rx_{c,t+1} = f_{c,t} - s_{c,t+1} = f_{c,t} - s_{c,t} - \Delta s_{c,t+1}.$$

We compute the currency excess return for portfolio p, $rx_{p,t+1}$, by taking the equally weighted average of the log currency excess returns in each portfolio. We rebalance portfolios at the end of each month. The $Carry^*$ factor is the return of P2 minus the return of P1. We report results with and without transaction costs, given the substantial turnover of currency strategies as exemplified in Burnside et al. (2007) and Menkhoff et al. (2012a). Net returns subtract, each month, the quoted forward half-spread of Subsection 2.1 of each currency times its turnover. Portfolio turnover[4] is computed as

$$TO_{p,t} = \sum_{c=1}^{N} \left| w_{c,t} - w_{c,t-1} \frac{1 + rx^l_{c,t-1}}{1 + rx^l_{p,t-1}} \right|,$$

where w_c is the weight of currency c in the portfolio p.

2.4. Benchmarks and sample

We compare $carry^*$ to two benchmark carry strategies, which differ in terms of data requirements. Carry, following Lustig et al. (2011), sorts currencies on short-term interest rate differentials, measured by the 1-month forward premium, $f_{c,t} - s_{c,t}$. Static carry, following Hassan and Mano (2019), sorts currencies on their average forward premium over the initial four-year window. All three strategies use the same median split and equal weights, and the dollar factor is the average excess return of all currencies. To ensure comparability, all three strategies begin at the same date and keep the same group of countries, determined by the most demanding data requirement. Therefore, all strategies begin in October 1987, following the four-year initialization window of static carry, with all countries except Australia, New Zealand and Sweden. Australia and New Zealand enter all portfolios in December 1988, reflecting their FX data start date of December 1984 combined with the four-year initialization window of static carry. Sweden enters all portfolios in January 1989, two years after its 10-year yield begins, so that the alternative versions of the long-run rate, some of which require two years of yields, share the same

[4]We use linear returns, $rx^l_{c,t}$, instead of log-returns, $rx_{c,t}$, when computing portfolio turnover. For most cases, the difference between $rx^l_{c,t}$ and $rx_{c,t}$ is negligible.

sample. At the euro's introduction in January 1999, the Deutsche mark exits all portfolios and the euro enters. For static carry, the euro inherits the Deutsche mark's initial forward premium ranking, preserving continuity of the sorting variable across the currency union transition. The common sample has 449 months and 3,998 currency-months, from October 1987 to February 2025.

3. Carry* and the cross-section of currency returns

In this section we analyze the returns of *carry** and its ability to price the cross-section of currency returns. First, we establish that *carry** is a genuinely distinct signal from carry, as it ranks currencies differently from the forward premium while being far more persistent. Second, we highlight the predictive ability of *carry** through univariate portfolio sorting. Third, we show that *carry** outperforms carry and static carry, and we use Fama-MacBeth regressions on two sets of G10 test assets and spanning regressions to show an asymmetric relationship in which *carry** subsumes the pricing information of both benchmarks, while retaining independent variation that neither can explain. Fourth, we examine the stability of *carry** across subperiods and states. Finally, we show that its premium is not explained by other currency factors documented in the literature.

3.1. Carry* is distinct from carry

The forward premium, $f_{c,t} - s_{c,t}$, is the standard sorting signal underlying traditional carry strategies and closely approximates the short-term interest rate differential via covered interest parity. Before examining the pricing properties of *carry**, it is instructive to assess the extent to which our signal overlaps with the carry signal. If the long-run differential ranked currencies in nearly the same way as the forward premium, *carry** would add little beyond a reparametrization of carry. At the other extreme, if it barely changed over time, it would be close to the static ranking of Hassan and Mano (2019).

Table 1 compares *carry**, carry and static carry along these two dimensions, together with a signal sorted on the 10-year yield differential itself. We measure persistence by the first-order autocorrelation of each signal, averaged across currencies, and by rank stability, the average Spearman correlation between the rankings of consecutive months. We measure the overlap with carry by the average cross-sectional rank correlation with the forward premium and by the share of currency-months in which a currency sits in the same portfolio as under carry.

The results show that *carry** lies between the two benchmarks. It is far more persistent than carry, with a first-order autocorrelation of 0.91 against 0.53 and a rank stability

of 0.94 against 0.68. At the same time, it is not a static ranking, as countries change positions over time as their long-run rates evolve. Its rank correlation with the forward premium is 0.55, lower than that of static carry, 0.60, and of the 10-year yield, 0.71. As a result, $carry^*$ holds a currency on the same side as carry in 62% of currency-months, so in more than a third of currency-months the two strategies take opposite positions. The higher persistence also translates into lower trading, as $carry^*$ turns over 4.82 times a year, against 10.89 for carry. The comparison with the 10-year yield shows that $carry^*$ is not simply a sort on long-term interest rates. The 10-year yield differential is even more persistent, but it ranks currencies closer to the forward premium and shares the side of carry in 70% of currency-months. By removing the part of the long rate not explained by inflation and growth, the fair value moves the signal further away from carry. This establishes that $carry^*$ is a genuinely distinct signal rather than a repackaged version of carry or of the long rate, which motivates examining its pricing properties in the following sections.

Table 1: Persistence of the signals and overlap with Carry. $Carry^*$ sorts currencies on their long-run rate differential to the US, the real-time fair value of the 10-year yield on inflation and GDP growth. 10-year yield sorts on the yield differential itself. AC(1) is the mean autocorrelation across currencies, rank stability the mean Spearman correlation of consecutive monthly rankings, and the rank correlation with the forward premium the mean monthly cross-sectional Spearman correlation. Same side as Carry is the share of currency-months in the same leg as under Carry. Turnover is annual and two-way. October 1987 to February 2025.

Signal	AC(1)	Rank stability	Rank correlation with the forward premium	Same side as Carry	Turnover
Carry*	0.91	0.94	0.55	0.62	4.82
10-year yield	0.98	0.98	0.71	0.70	2.56
Carry	0.53	0.68	1.00	1.00	10.89
Static Carry		1.00	0.60	0.62	0.65

3.2. Carry* commands a positive excess return

We use the standard methodology of univariate portfolio sorting to assess the unconditional pricing ability of the $carry^*$ signal. Table 2 shows the summary statistics of the one-month ahead excess returns of the two median portfolios, P1 and P2, as well as the return on the self-financing portfolio that buys currencies above the median long-run differential and shorts those below, P2 − P1.

The results in Table 2 indicate a strong positive relationship between the $carry^*$ measure and future excess returns. The annualized average return spread between the high (P2) and low (P1) portfolios is 3.52%, and this spread is statistically significant with

Table 2: Portfolios sorted on $Carry^*$. $Carry^*$ sorts currencies on their long-run rate differential to the US, the real-time fair value of the 10-year yield on inflation and GDP growth. P1 holds the currencies below the monthly cross-sectional median and P2 those above, equally weighted, and P2 − P1 is the $Carry^*$ factor. Means and standard deviations are annualized in %. The t-statistic uses Newey-West standard errors (one lag). October 1987 to February 2025.

	P1	P2	P2 − P1
Mean (%)	−1.31	2.21	3.52
Std. Dev. (%)	7.93	8.98	6.16
Sharpe ratio	−0.17	0.25	0.57
t-statistic	−0.99	1.44	3.45

a t-statistic of 3.45. The performance of $carry^*$ cannot be attributed to a single portfolio leg. In fact, the short leg delivers a negative average return of −1.31% while the long leg delivers a positive average return of 2.21%, so both legs contribute to the spread, although neither is significant on its own. The spread portfolio also has lower volatility than either P1 or P2, 6.16% against 7.93% and 8.98%, again highlighting the usefulness of the long-short strategy. Consequently, the Sharpe ratio of the high minus low portfolio, 0.57, is higher than that of either of its components, −0.17 and 0.25. The high minus low portfolio is the $carry^*$ factor, whose performance we compare to the benchmark carry strategies in the next subsection.

3.3. Carry* outperforms and subsumes benchmark carry strategies

We next ask whether the $carry^*$ factor improves upon the traditional carry factor of Lustig et al. (2011) or the static rank approach of Hassan and Mano (2019). To this end, we compare their performance as trading strategies, evaluate their risk-pricing ability through Fama-MacBeth regressions and analyze the significance of abnormal returns through time-series spanning regressions. Our results highlight that $carry^*$ outperforms carry and static carry both as a trading strategy and as a priced risk factor in currency markets.

Table 3 shows the summary statistics of the one-month ahead excess returns of $carry^*$, carry and static carry, built as in **??**. The trading strategy based on $carry^*$ delivers superior returns compared to both benchmarks. $Carry^*$ yields a higher average excess return, 3.52% against 2.18% for carry and 1.84% for static carry, with a similar volatility, and therefore a higher Sharpe ratio, 0.57 against 0.35 and 0.30. The gap with carry widens when accounting for transaction costs, as $carry^*$ turns over less than half as often as carry. Net of quoted bid-ask spreads, the Sharpe ratio of $carry^*$ is 0.54, against 0.29 for carry and 0.30 for static carry, whose ranking barely changes over time. The Sortino ratio, which

penalizes only downside volatility, further confirms the superior risk-return relationship of $carry^*$, showing that it generates more return per unit of downside risk. The Omega ratio, which measures the ratio of the total magnitude of gains to the total magnitude of losses relative to a zero threshold, is also higher for $carry^*$, indicating that it generates proportionally larger gains relative to losses than either benchmark. Moreover, the return distribution of $carry^*$ exhibits a less negative skewness than carry, -0.43 against -0.64, which is an important improvement given the well-documented downside risks associated with carry trades (Brunnermeier et al., 2008), and is similar to that of static carry. $Carry^*$ also exhibits the lowest excess kurtosis of the three strategies, which indicates less extreme returns. Its maximum drawdown, however, is deeper than that of carry, 23% against 18%, and close to the 24% of static carry. Taken together, these results highlight that isolating the long-run component of interest rate differentials not only enhances profitability but also improves the risk-return tradeoff of carry strategies.

Table 3: Performance of $Carry^*$, Carry and Static Carry. $Carry^*$ sorts currencies on their long-run rate differential to the US, the real-time fair value of the 10-year yield on inflation and GDP growth. Carry sorts on the forward premium and Static Carry on its mean over each currency's first four years. Each strategy is long the currencies above the monthly cross-sectional median of its signal and short those below. Means and standard deviations are annualized in %, and the t-statistic uses Newey-West standard errors (one lag). The net Sharpe ratio subtracts quoted bid-ask half-spreads on turnover (Table A3). Kurtosis is excess kurtosis. Cumulative return and maximum drawdown are in decimals. October 1987 to February 2025.

	Carry*	Carry	Static Carry
Mean (%)	3.52	2.18	1.84
Std. Dev. (%)	6.16	6.21	6.04
Sharpe ratio	0.57	0.35	0.30
t-statistic	3.45	2.18	1.84
Turnover	4.82	10.89	0.65
Sharpe ratio (net)	0.54	0.29	0.30
Sortino	0.82	0.47	0.44
Omega	1.53	1.31	1.25
Skewness	−0.43	−0.64	−0.41
Kurtosis	1.19	1.89	1.38
Cumulative return	2.47	1.10	0.86
Maximum drawdown	−0.23	−0.18	−0.24

Our results are also economically significant. Figure 2 shows the cumulative value of a one-dollar initial investment in the three strategies. Our $carry^*$ strategy yields a cumulative return of 247% over the sample period, higher than 110% from carry and 86% from static carry. The advantage of $carry^*$ is larger in the first half of the sample, a pattern we examine in the subperiod analysis below.

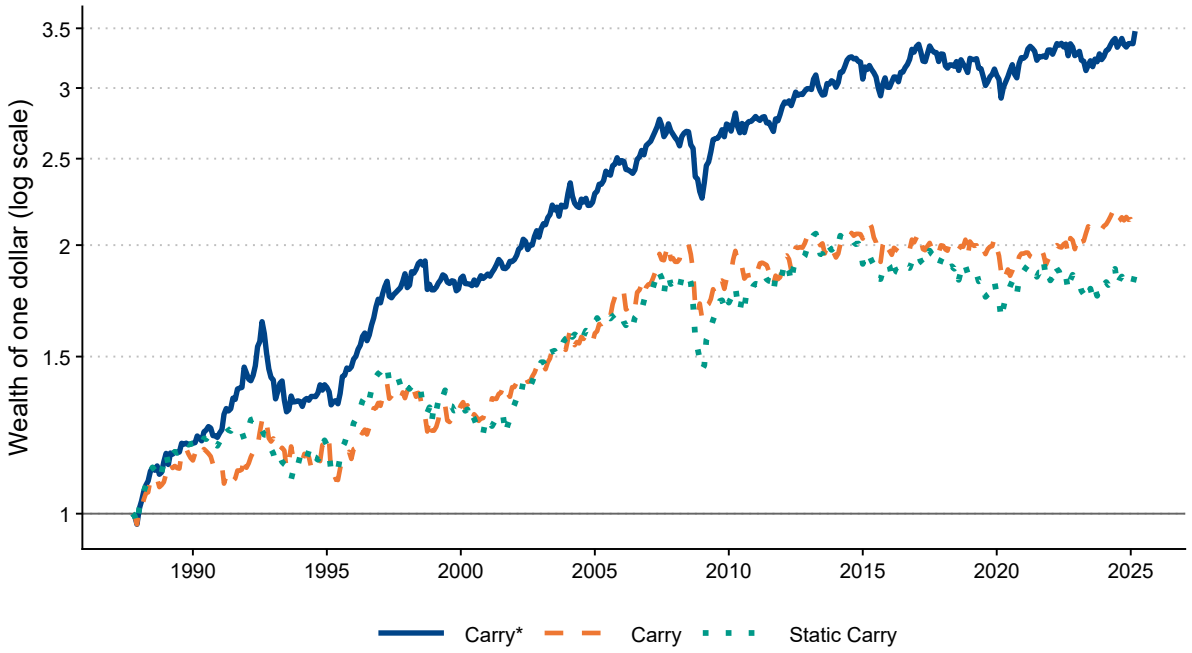


Figure 2: Cumulative excess returns of long-short, equal-weighted currency portfolios sorted on *carry**, carry and static carry. Each month, currencies are sorted by the previous month's value of the characteristic. We go long those above the cross-sectional median and short those below, rebalancing monthly. Each line is the wealth of one dollar invested in October 1987, on a log scale. Returns are gross of transaction costs. Sample October 1987 to February 2025.

To formally evaluate whether *carry** contains independent pricing information beyond established currency return factors, we employ the two-pass methodology of Fama and MacBeth (1973). Since Lustig and Verdelhan (2007), this approach has become standard in currency asset pricing, directly linking returns to risk factors through the stochastic discount factor (SDF). In the first pass, we estimate β_n from time-series regressions of each test asset's excess return on the factors f_t:

$$rx_{n,t} = \alpha_n + f_t\beta_n^\top + \epsilon_{nt}, \quad n = 1, \ldots, N, \quad t = 1, \ldots, T,$$

with α_n capturing the risk-adjusted average excess return. In the second step, expected returns are proxied by average realized returns, $\overline{rx}_n = \frac{1}{T}\sum_{t=1}^{T} rx_{n,t}$, and regressed on the estimated betas (factor loadings) without an intercept:

$$\overline{rx}_n = \hat{\beta}_n\lambda^\top + a_n, \quad n = 1, \ldots, N,$$

where a_n denotes the pricing error and $\hat{\beta}_n\lambda^\top$ the model-implied risk premium. The risk

prices are estimated as

$$\hat{\lambda} = (\hat{\beta}^\top \hat{\beta})^{-1} \hat{\beta}^\top \overline{rx},$$

with $\hat{\beta}$ the $N \times K$ matrix of estimated factor loadings and $\overline{rx}$ the $N \times 1$ vector of average excess returns. We report Shanken (1992) standard errors, which correct for the estimation error in the first-pass betas.

We consider four factors: $carry^*$, carry, static carry and the dollar factor of Lustig et al. (2011). The $carry^*$ factor is the return on the high minus low portfolio of Subsection 3.2, and the carry and static carry factors are built in the same way, as described in **??**. The dollar factor is the excess return on the equal-weighted portfolio of all currencies and enters every specification. Since the dollar factor plays the role of a common intercept for currency portfolios, the second pass has no intercept. In addition, the traded factors other than the dollar factor are added to the test assets, so that each factor is required to price itself. We use two sets of G10 test assets. The first consists of our own 27 rank portfolios, in which each of the nine currencies is ranked on $carry^*$, the forward premium and static carry. The second consists of the 45 G10 quintile portfolios of Della Corte and Riddiough (2025), sorted on carry, value, momentum over one, three, six and twelve months, volatility, skewness and kurtosis. The second set captures distinct factor structures in currency returns that are not built from our signals, and therefore provides a demanding evaluation of whether $carry^*$ is able to explain the cross-sectional variation in currency excess returns beyond the benchmark factors.

Table 4 reports estimated annualized prices of risk, along with their associated Shanken-corrected standard errors. The first column in each panel shows that the $carry^*$ factor alone commands a positive and significant annualized price of risk, 2.74% on our own portfolios and 3.47% on the portfolios of Della Corte and Riddiough (2025). This implies that assets with higher first-stage factor betas with respect to $carry^*$ are compensated with higher expected returns, consistent with the portfolio-sorting evidence in Subsection 3.2. Carry and static carry alone also command positive prices of risk, but with lower significance and lower explanatory power. The cross-sectional R^2 is 0.48 for $carry^*$ against 0.35 for carry and 0.28 for static carry in Panel A, and 0.58 against 0.42 and 0.40 in Panel B.

Columns 4–6 report results from specifications that include $carry^*$ jointly with the benchmarks. When $carry^*$ and carry are included together, $carry^*$ remains significant at the 1% level in both panels, while carry loses its significance. A similar pattern emerges when $carry^*$ is included alongside static carry. When all three factors are included, the risk price of $carry^*$ is 3.02% and 3.73%, significant at the 1% level, while carry and static carry are at most significant at the 10% level, carry in Panel A and static carry in Panel

B. The dollar factor is insignificant in every specification. The R^2 values further reinforce this picture, as $carry^*$ alone achieves higher explanatory power than either benchmark alone in both panels, and adding the benchmarks raises it only slightly.

Overall, the Fama-MacBeth regressions confirm that $carry^*$ earns a positive and significant risk premium in the cross-section of currency returns, both on portfolios built from our signals and on portfolios that capture distinct factor structures, remaining statistically and economically significant even after accounting for alternative carry measures and the dollar factor.

Table 4: Fama-MacBeth risk prices on G10 test assets. Second-stage risk prices in % a year, with Shanken (1992) standard errors in parentheses. The second pass has no intercept and adds the traded factors other than Dollar to the test assets. Panel A uses our 27 rank portfolios of the nine currencies on $Carry^*$, the forward premium and Static Carry (January 1989 to February 2025), and Panel B the 45 G10 quintile portfolios of Della Corte and Riddiough (2025), DCR (October 1987 to February 2025). Dollar is the mean excess return of all currencies. MAPE is the mean absolute pricing error. Returns are log returns. *, ** and *** denote significance at 10, 5 and 1%.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Own G10 test assets (27 portfolios)						
Carry*	2.74**			2.87***	3.05***	3.02***
	(1.15)			(1.07)	(1.06)	(1.06)
Carry		2.27*		1.51		1.76*
		(1.16)		(1.11)		(1.06)
Static Carry			1.72		0.94	0.91
			(1.10)		(1.08)	(1.07)
Dollar	−0.21	−0.19	−0.19	−0.21	−0.20	−0.20
	(1.30)	(1.30)	(1.30)	(1.30)	(1.30)	(1.30)
R^2	0.48	0.35	0.28	0.50	0.50	0.52
MAPE (%)	1.11	1.19	1.30	1.11	1.07	1.06
Panel B: DCR G10 test assets (45 portfolios)						
Carry*	3.47***			3.78***	3.45***	3.73***
	(1.15)			(1.14)	(1.21)	(1.17)
Carry		2.85**		1.64		1.56
		(1.17)		(1.11)		(1.11)
Static Carry			2.53**		1.82	1.92*
			(1.10)		(1.12)	(1.13)
Dollar	0.03	0.06	0.07	0.02	0.03	0.02
	(1.30)	(1.30)	(1.30)	(1.30)	(1.30)	(1.30)
R^2	0.58	0.42	0.40	0.61	0.60	0.63
MAPE (%)	0.60	0.63	0.67	0.60	0.60	0.60

While the Fama-MacBeth regressions establish that $carry^*$ commands a significant price

of risk in the cross-section, they do not directly address whether *carry** contains return-relevant information beyond what carry and static carry already capture, or conversely whether those benchmarks are redundant once *carry** is included. We examine this through time-series spanning regressions, where a significant intercept indicates that the dependent factor earns abnormal returns unexplained by the regressors, while an insignificant intercept is evidence of subsumption. All regressions include the dollar factor.

Table 5 shows the results. Columns (1) to (3) show that *carry** earns a statistically significant abnormal return of 2.16% per year after controlling for carry, 2.29% after controlling for static carry and 1.93% after controlling for both, all significant at the 1% level. This indicates that *carry** captures risk premia not fully reflected in existing carry measures. In contrast, columns (4) to (6) show that the intercepts of carry and static carry are economically small, between -0.10% and -0.05% per year, and statistically indistinguishable from zero once *carry** is included as a regressor. Notably, in column (5) static carry loads positively and significantly on the dollar factor, suggesting that part of its return is dollar exposure that *carry** does not share. Taken together, these results indicate an asymmetric spanning relationship, in which *carry** subsumes the pricing information of both carry and static carry, while retaining independent variation that neither can explain. The premium is also cross-sectional, since a time-series version that holds each currency long when its long-run rate is above the US long-run rate is long the dollar in most months, and the part of its return that comes from this dollar position is not statistically significant.[5]

[5] In Appendix H, the time-series version of *carry** earns a Sharpe ratio of 0.47, below the 0.57 of the cross-sectional strategy, and the mean long-run differential has only some power to predict the dollar, with an IVX p-value of 0.041 at one month.

Table 5: Spanning regressions. Each column regresses the factor named above it on the factors with a coefficient in that column, always with Dollar. The intercept is the alpha in % a year. Newey-West standard errors (12 lags) in parentheses. *, ** and *** denote significance at 10, 5 and 1%. October 1987 to February 2025.

	Carry*			Carry	Static Carry	Carry
	(1)	(2)	(3)	(4)	(5)	(6)
Intercept (α, %)	2.16***	2.29***	1.93***	−0.09	−0.10	−0.05
	(0.73)	(0.64)	(0.61)	(0.63)	(0.53)	(0.56)
Carry*				0.63***	0.54***	0.41***
				(0.05)	(0.09)	(0.07)
Carry	0.60***		0.40***			
	(0.05)		(0.09)			
Static Carry		0.63***	0.36***			0.40***
		(0.07)	(0.10)			(0.08)
Dollar	0.08*	−0.02	0.00	0.03	0.20***	−0.05
	(0.05)	(0.07)	(0.06)	(0.06)	(0.05)	(0.05)
Adj. R^2	0.40	0.36	0.47	0.39	0.43	0.48

3.4. Stability across subperiods and states

A natural concern with full-sample results is that the performance of *carry** is concentrated in a particular period or in particular states of the economy. To address this, we split the sample into subperiods and the months into states, and we compare *carry** with carry and static carry in each of them.

Panel A of Table 6 reports the Sharpe ratios of *carry** and carry in two halves of the sample, split at 2006, and in three periods of similar length, together with the abnormal return of *carry** over carry, static carry and the dollar factor. *Carry** delivers a higher Sharpe ratio than carry in every subperiod. Its performance is strongest before 2006, with a Sharpe ratio of 0.82 against 0.53 for carry and an abnormal return of 2.95% per year with a t-statistic of 2.44. After 2006, both strategies earn less. Even so, the Sharpe ratio of *carry** is 0.32 against 0.18 for carry, and its abnormal return of 1.08% per year remains positive with a t-statistic of 1.79. The three shorter periods show the same pattern, as the abnormal return of *carry** is positive in each of them, ranging from 1.40% to 3.33% per year. Panel B of Table 6 splits the months into high and low states by the median or the sign of each state variable, measured at the time of portfolio formation. The state variables capture inflation in the US and in the G10, global risk through the level of the VIX and FX volatility, the trend of the dollar, the direction of long-term yields and the direction of US monetary policy. For each state, we report the mean return and the abnormal return

of $carry^*$, and the mean returns of carry and static carry, together with the t-statistic of the difference between the high and low states.

The results show that the performance of $carry^*$ is not driven by a specific state of the economy. Its mean return is positive in every state, ranging from 1.85% to 4.79% per year, and its abnormal return over the benchmarks is also positive in every state. More importantly, $carry^*$ earns a higher mean return than both carry and static carry in all 14 states. The differences between high and low states are not statistically significant at the 5% level for any of the three strategies, although some of them are economically sizable but imprecisely estimated, such as the abnormal return of $carry^*$ across FX volatility states, 0.79% against 3.20% per year. The absence of significant state dependence indicates that the premium of $carry^*$ is not a disguised bet on a particular regime, such as periods of high inflation, calm markets or a weak dollar. Since the benchmarks do not vary significantly across states either, the state splits do not separate the three strategies, and what separates them is that $carry^*$ earns more in every state. $Carry^*$ tends to earn more when FX volatility is low, when long-term yields fall and when the Fed eases. The benchmarks share this tendency for yields and monetary policy, while they earn more when FX volatility is high. Taken together, these results indicate that the advantage of $carry^*$ lies in the level of its returns rather than in its timing across states of the economy.

Table 6: $Carry^*$ in sub-periods and states. Panel A gives the Sharpe ratios of $Carry^*$ and Carry in each sub-period and the alpha of $Carry^*$ over Dollar, Carry and Static Carry in % a year (Newey-West t-statistic). Panel B splits the months by the median or the sign of each state variable, known at formation. Each cell gives the value in the high and the low state, high / low, for the mean return of $Carry^*$, its alpha over Dollar, Carry and Static Carry, and the mean return of Carry and of Static Carry, with the Newey-West t-statistic of the difference, high minus low, in parentheses. October 1987 to February 2025.

Panel A: Sub-periods	1987–2005	2006–2025	1987–1999	2000–2012	2013–2025
Carry, Sharpe ratio	0.53	0.18	0.37	0.56	0.07
Carry*, Sharpe ratio	0.82	0.32	0.75	0.68	0.24
Carry*, α (%)	2.95	1.08	3.33	1.40	1.53
	(2.44)	(1.79)	(2.23)	(2.16)	(1.86)

Panel B: States	Carry*, mean (% a year)	Carry*, α (%)	Carry, mean (% a year)	Static Carry, mean (% a year)
US inflation	3.34 / 3.71	2.13 / 1.62	2.09 / 2.27	0.79 / 2.87
	(−0.16)	(0.36)	(−0.10)	(−0.88)
G10 inflation	4.13 / 2.91	2.56 / 1.30	2.22 / 2.14	1.80 / 1.87
	(0.64)	(1.01)	(0.04)	(−0.04)
VIX level	3.89 / 2.55	2.20 / 1.59	2.36 / 1.48	2.26 / 0.59
	(0.65)	(0.56)	(0.48)	(0.74)
FX volatility (trailing 12m)	2.97 / 4.08	0.79 / 3.20	2.82 / 1.55	2.90 / 0.77
	(−0.55)	(−1.74)	(0.65)	(1.14)
Dollar weak (high) vs strong	4.49 / 2.44	1.83 / 1.70	3.33 / 0.90	3.05 / 0.48
	(1.26)	(0.10)	(1.53)	(1.56)
G10 10y yield rising (high) vs falling	1.85 / 4.56	1.55 / 2.25	1.47 / 2.62	−0.26 / 3.14
	(−1.42)	(−0.58)	(−0.62)	(−1.73)
Fed tightening (high) vs easing	2.38 / 4.79	1.26 / 2.67	2.08 / 2.29	0.77 / 3.01
	(−1.21)	(−1.17)	(−0.11)	(−1.08)

3.5. Carry* and other currency factors

So far we have compared *carry** with carry and static carry. However, the literature has documented several other sources of currency excess returns, and a natural concern is that *carry** simply repackages one of them. We consider currency value (Asness et al., 2013; Menkhoff et al., 2017) and currency momentum (Menkhoff et al., 2012b), a term spread factor that sorts currencies on the slope of the yield curve (Ang and Chen, 2010), a global imbalance factor that is long the currencies of net debtor countries (Della Corte et al., 2016), a size factor that is long the currencies of small economies (Hassan, 2013) and the commodity factor of Ready et al. (2017). We build the value, momentum, term spread, global imbalance and size factors with the same median split as *carry** on our nine currencies, using the net foreign assets and GDP data of Lane and Milesi-Ferretti (2018) for the last two. The commodity factor is taken from the data of Ready et al. (2017), available from WRDS, which end in 2013.

Panel A of Table 7 reports the abnormal return of *carry** after controlling for the dollar factor and each group of factors, first alone and then together with carry and static carry. The abnormal return of *carry** remains positive and statistically significant in every specification. It is 3.07% per year after controlling for value and momentum, 3.26% after controlling for the term spread, 2.50% after controlling for global imbalance and size, and 3.45% after controlling for the commodity factor. When carry and static carry are added, the abnormal return of *carry** ranges from 1.93% to 2.57% per year, with t-statistics above 3. These results indicate that *carry** captures a source of currency returns that is not explained by the other factors documented in the literature.[6]

Panel B of Table 7 looks at the other factors themselves, reporting their mean return, their Sharpe ratio and their abnormal return after controlling for the dollar factor and *carry**. Value delivers a Sharpe ratio of 0.40 and retains a significant abnormal return of 2.19% per year, so value and *carry** capture distinct information. Momentum earns little in our sample of G10 currencies at any of the three horizons. The term spread factor of Ang and Chen (2010) earns −0.30% per year, with a Sharpe ratio of −0.05. This contrast is important for the interpretation of *carry**. Both signals use information from long-term yields, but sorting on the slope of the yield curve delivers no premium, while sorting on the long-run rate does, which separates the long-run rate from the term spread. Finally, the global imbalance and size factors earn positive average returns, but their abnormal

[6] Appendix E shows that none of the term spread, global imbalance, size and commodity factors earns a significant abnormal return over *carry**, and that countries with high long-run rates tend to be net debtors, small and commodity exporters, although less so than countries with high 10-year yields.

returns are small and statistically insignificant once $carry^*$ is included, so $carry^*$ spans them. The commodity factor, available from 1988 to 2013, delivers the highest Sharpe ratio, 0.53, but its abnormal return over $carry^*$ is not significant either.

Table 7: $Carry^*$ and other currency factors. Panel A gives the alpha of $Carry^*$ over Dollar and the factors in each row, alone and with Carry and Static Carry. Panel B gives the mean and Sharpe ratio of each factor and its alpha over Dollar and $Carry^*$. All factors except commodity are built with the median split of $Carry^*$ on our nine currencies. Value sorts on the 5-year real depreciation against the US dollar (Asness et al., 2013; Menkhoff et al., 2017), momentum on the excess return of the past 1, 3 or 12 months (Menkhoff et al., 2012b) and the term spread factor on the 10-year yield minus the short rate (Ang and Chen, 2010). Global imbalance is long the currencies of net debtors (Della Corte et al., 2016), with net foreign assets the net international investment position excluding gold over GDP, and size is long the currencies of small economies (Hassan, 2013), with size the log share of world GDP, both from the External Wealth of Nations data of Lane and Milesi-Ferretti (2018). Commodity is the factor of Ready et al. (2017), taken from their data on WRDS, which end in 2013. Alphas in % a year. Newey-West t-statistics (12 lags) in parentheses. October 1987 to February 2025.

Panel A: Alpha of Carry* over other factors

	Alpha (%)	Alpha with Carry and Static Carry (%)
Value and momentum	3.07	1.96
	(3.86)	(3.44)
Term spread	3.26	1.93
	(3.56)	(3.13)
Global imbalance and size	2.50	2.06
	(3.12)	(3.49)
Commodity (1988–2013)	3.45	2.57
	(2.87)	(3.34)

Panel B: The other factors

	Mean (%)	Sharpe ratio	Alpha over Dollar and Carry* (%)
Value	2.46	0.40	2.19
			(2.31)
Momentum, 1 month	0.67	0.10	0.48
			(0.46)
Momentum, 3 months	−0.82	−0.13	−1.26
			(−1.25)
Momentum, 12 months	0.16	0.02	−0.39
			(−0.51)
Term spread	−0.30	−0.05	0.25
			(0.26)
Global imbalance	1.87	0.29	0.20
			(0.26)
Size	1.31	0.26	0.59
			(1.03)
Commodity (1988–2013)	4.61	0.53	1.83
			(1.19)

4. Understanding the premium

In this section we study where the premium of $carry^*$ comes from and what risks it carries. First, we split the return of $carry^*$ into the interest differential it earns and the currency movement it experiences, use uncovered interest parity regressions to identify which part of the interest differential exchange rates offset, and split the long-run rate into its inflation and real parts. Second, we study the returns of $carry^*$ in bad times and its exposure to crash risk and oil risk.

4.1. Where the premium comes from

Having established that $carry^*$ earns a premium that carry, static carry and other currency factors cannot explain, we now ask where this premium comes from and which part of the long-run rate is priced. We first split the return of each strategy into the interest differential it earns and the currency movement it experiences, and into the part earned by holding currencies in fixed positions and the part earned by changing positions over time. We then use uncovered interest parity regressions to identify which part of the interest differential is offset by exchange rates. Finally, we split the 10-year yield along the Fisher equation into an inflation part and a real part, and we ask which of the two carries the premium.

Since $rx_{c,t+1} = (f_{c,t} - s_{c,t}) - \Delta s_{c,t+1}$, the log excess return of a long-short portfolio with weights $w_{c,t}$ splits exactly into two parts,

$$\sum_c w_{c,t}\, rx_{c,t+1} = \underbrace{\sum_c w_{c,t}\,(f_{c,t} - s_{c,t})}_{\text{interest differential}} \underbrace{- \sum_c w_{c,t}\,\Delta s_{c,t+1}}_{\text{currency movement}}, \tag{3}$$

where the weights are $1/N_t^H$ for the N_t^H currencies in the long portfolio and $-1/N_t^L$ for the N_t^L currencies in the short portfolio. The interest differential is the forward premium the strategy earns at formation, which by covered interest parity equals the short-term interest rate differential to the US. The currency movement is the gain or loss from the change in exchange rates over the following month, positive when the currencies held long appreciate against those held short. Under uncovered interest parity, the currency movement offsets the interest differential and the strategy earns nothing on average.

Columns 1–3 of Table 8 report the average of each part over the sample, annualized, together with Newey-West t-statistics for the currency movement and the total. The two parts add up to the mean log excess return, which is slightly lower than the mean excess return of Table 2 and Table 3. The split must use log returns, since only in logs

does the excess return separate exactly into the interest differential and the currency movement. The simple return of each currency exceeds its log return by about half its squared return, so the totals in column 3 fall short of the simple means by between 0.05 and 0.16 percentage points per year. The results show that the strategies earn their returns in different ways. Carry earns the largest interest differential, 4.09% per year, but loses 1.96% per year through currency movement, so that almost half of the differential is offset by exchange rates. Static carry earns an interest differential of 2.60% and loses 0.91% through currency movement. $Carry^*$, in contrast, earns a smaller interest differential than carry, 2.49% per year, but gains 0.98% per year through currency movement, for a total of 3.47%. The currency gain of $carry^*$ is not statistically significant on its own, but it differs sharply from the significant currency loss of carry. In other words, $carry^*$ selects currencies with a positive interest differential that exchange rates do not take away.

Figure 3 shows the cumulative value of each part over the sample, and Figure 4 and Figure 5 split them by subperiod and by currency. The currency movement of $carry^*$ is positive in three of the four subperiods and is the largest between 1996 and 2005, at 2.76% per year, while that of carry is negative in three of the four subperiods and reaches -5.47% per year before 1996. The currency movement of $carry^*$ is higher than that of carry in every subperiod. Across currencies, the currency gain of $carry^*$ comes mainly from the Australian dollar, the New Zealand dollar and the Norwegian krone, which it holds long most of the time. At the same time, it loses little through the Japanese yen and the Swiss franc, which it holds short in most months, while these two currencies cost carry 0.37% and 0.69% per year through currency movement. Since long-run differentials are persistent, one may ask whether $carry^*$ earns its premium simply by holding the same currencies long and short throughout the sample. Hassan and Mano (2019) show that most of the return of carry comes from such a static position, and Feunou et al. (2024) find the same for G9 currencies. To answer this question, we follow Hassan and Mano (2019) and split the mean log excess return of each strategy into a static and a dynamic part. Let $\bar{w}_c$ be the mean portfolio weight of currency c over the sample. The static part is the return of holding each currency at its mean weight in every month, $\sum_c \bar{w}_c \, rx_{c,t+1}$, and the dynamic part is the rest, $\sum_c (w_{c,t} - \bar{w}_c) \, rx_{c,t+1}$, the return from changing positions over time. The two parts add up to the return in Equation 3 in every month. Since the long and short weights sum to zero in every month, the dollar trade of Hassan and Mano (2019) is zero, and the dynamic part corresponds to their forward premium trade. We further split each part into the interest differential and the currency movement, as in columns 1–3. Because the mean weights use the whole sample, the split is descriptive.

Columns 4–9 of Table 8 report the results, with the static part in columns 4–6 and the dynamic part in columns 7–9, so that columns 6 and 9 add up to column 3. The return of $carry^*$ comes mostly from changing positions. Its dynamic part is 2.29% per year with a t-statistic of 3.34, two thirds of its total return (column 10), while its static part, 1.18% per year, is not statistically significant. Carry shows the opposite pattern, in line with Hassan and Mano (2019). More than half of its return is static, with a dynamic share of 42% in column 10, and its dynamic part, 0.89% per year, is not statistically significant. As expected, static carry earns almost nothing from changing positions, since its ranking is fixed by construction. The split of the dynamic part in columns 7 and 8 shows where the difference comes from. The dynamic trades of carry earn a large interest differential, 2.27% per year, but lose 1.38% per year through currency movement, with a t-statistic of -2.11. The dynamic trades of $carry^*$ earn a smaller interest differential, 0.78% per year, and gain 1.51% per year through currency movement, with a t-statistic of 2.16.

These results show that $carry^*$ is not a repackaged static carry strategy. As argued in the introduction, it sits between carry and static carry. Like static carry, it ranks currencies on persistent differences, but unlike static carry, it updates positions as long-run rates change, and this updating is where most of its premium comes from. When carry changes positions, the exchange rate moves against it, and when $carry^*$ changes positions, the exchange rate moves in its favor. The uncovered interest parity regressions below show why.

To identify which part of the interest differential exchange rates offset, we estimate uncovered interest parity regressions in the spirit of Fama (1984). We regress the depreciation of each currency against the US dollar over the next month on its forward premium at the end of the month,

$$\Delta s_{c,t+1} = a + \beta \left(f_{c,t} - s_{c,t} \right) + \varepsilon_{c,t+1}, \tag{4}$$

pooling the nine currencies, with both variables annualized in percent. Under uncovered interest parity, $\beta = 1$, so that the currency depreciates by the interest differential. A slope below one means the currency loses less than the differential, and a negative slope means that it appreciates. To separate the long-run part of the forward premium from the rest, we first project the forward premium on the long-run differential, $f_{c,t} - s_{c,t} = k_0 + k_1\, carry^*_{c,t} + u_{c,t}$, over the pooled sample. We define the long-run part as $\hat{k}_1\, carry^*_{c,t}$ and the rest as the forward premium minus the long-run part, and we enter both in Equation 4 in place of the forward premium, each with its own slope. We estimate the regressions in two ways. The pooled version uses all the variation in the data. The month effects version

Table 8: Where the return comes from. Columns 1–3 split the mean log excess return of each strategy into the interest differential earned (portfolio weights times log forward premia) and the currency movement (the rest). Columns 4–9 split the same return into a static part (columns 4–6) and a dynamic part (columns 7–9) following Hassan and Mano (2019), each with the same two components, so columns 6 and 9 add up to column 3. The static part holds each currency at its mean weight over the sample, $\bar{w}_c$, so its monthly return is $\sum_c \bar{w}_c\, rx_{c,t+1}$. The dynamic part is the rest, $\sum_c (w_{c,t} - \bar{w}_c)\, rx_{c,t+1}$, the return from changing positions over time. Column 10 is the dynamic part as a percentage of the total, column 9 divided by column 3. The long and short weights sum to zero in every month, so the dollar trade of Hassan and Mano (2019) is zero. The mean weights use the whole sample, so the split is descriptive. The Deutsche mark and the euro count as one currency. Means in % a year with Newey-West t-statistics in parentheses. The interest differential is known when the portfolio is formed, so it has no t-statistic. October 1987 to February 2025.

	Total return			Static part			Dynamic part			
	Interest differential (1)	Currency movement (2)	Total (3)	Interest differential (4)	Currency movement (5)	Total (6)	Interest differential (7)	Currency movement (8)	Total (9)	Dynamic share (%) (10)
Carry*	2.49	0.98	3.47	1.71	−0.52	1.18	0.78	1.51	2.29	66
		(1.01)	(3.46)		(−0.79)	(1.78)		(2.16)	(3.34)	
Carry	4.09	−1.96	2.13	1.83	−0.58	1.24	2.27	−1.38	0.89	42
		(−2.02)	(2.33)		(−0.86)	(1.81)		(−2.11)	(1.50)	
Static Carry	2.60	−0.91	1.68	2.53	−1.11	1.42	0.07	0.19	0.26	16
		(−0.92)	(1.62)		(−1.09)	(1.39)		(1.33)	(1.21)	

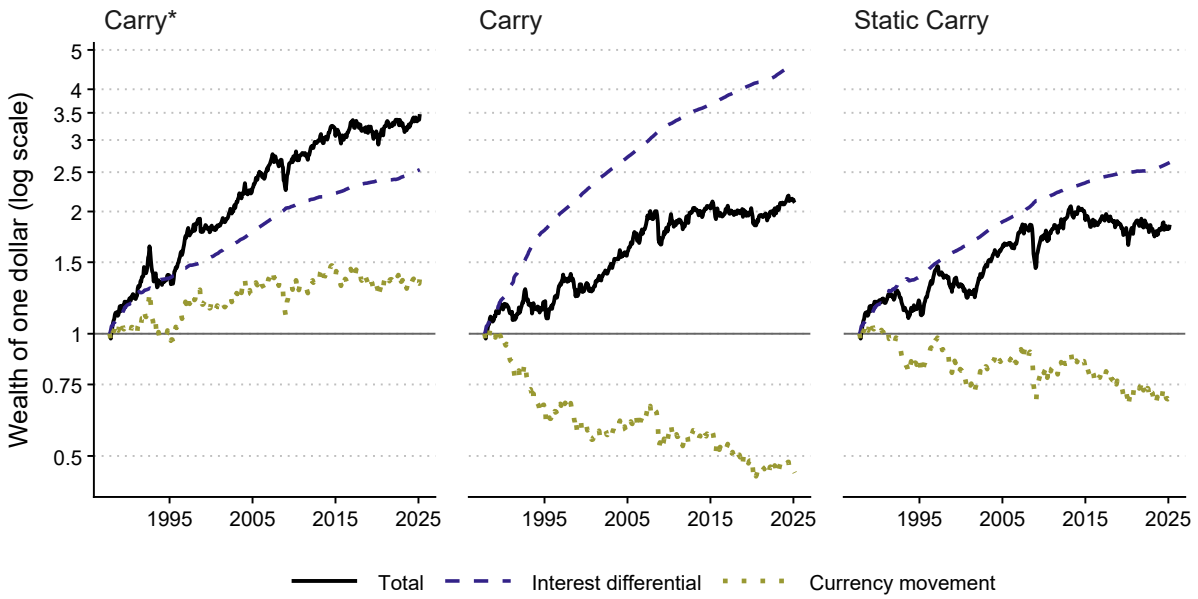


Figure 3: Interest differential and currency movement of the long-short portfolios sorted on *carry**, carry and static carry. In each panel, the solid line is the wealth of one dollar invested in the strategy, as in Figure 2. The dashed line is the wealth from the interest differential alone (the portfolio weights times the log forward premia), and the dotted line the wealth from the currency movement, the rest of the excess return, so that the two multiply to the total. Log scale. Sample October 1987 to February 2025.

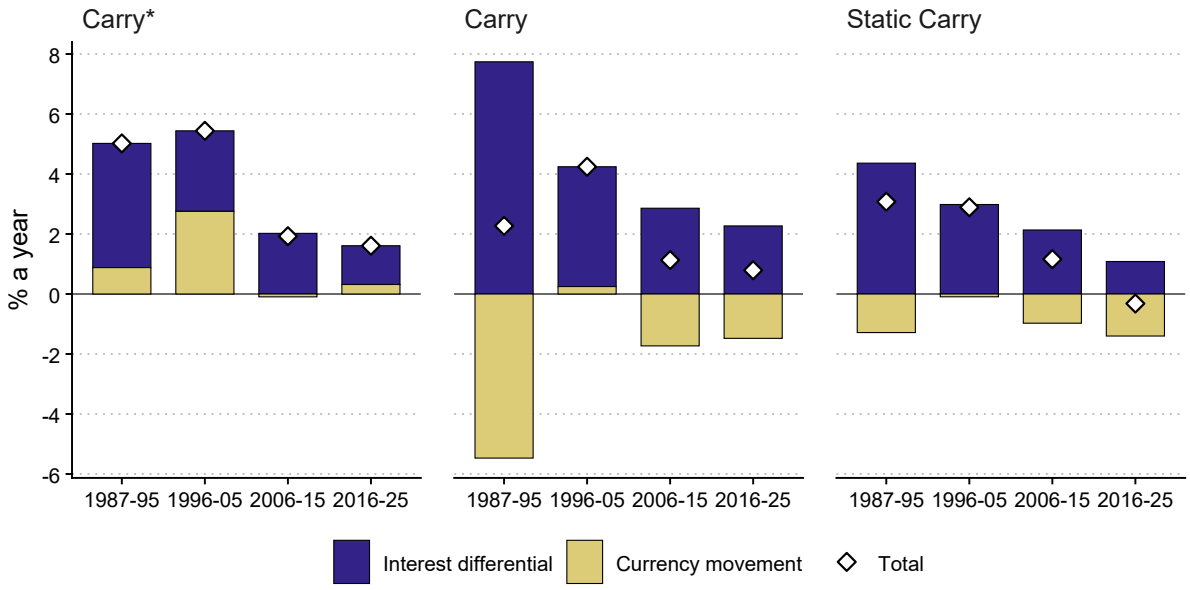


Figure 4: Interest differential and currency movement of the long-short returns by sub-period. Bars split the mean log excess return of *carry**, carry and static carry into the interest differential earned and the currency movement. Diamonds mark the total. Sub-periods are October 1987 to December 1995, 1996 to 2005, 2006 to 2015 and 2016 to February 2025. In % a year.

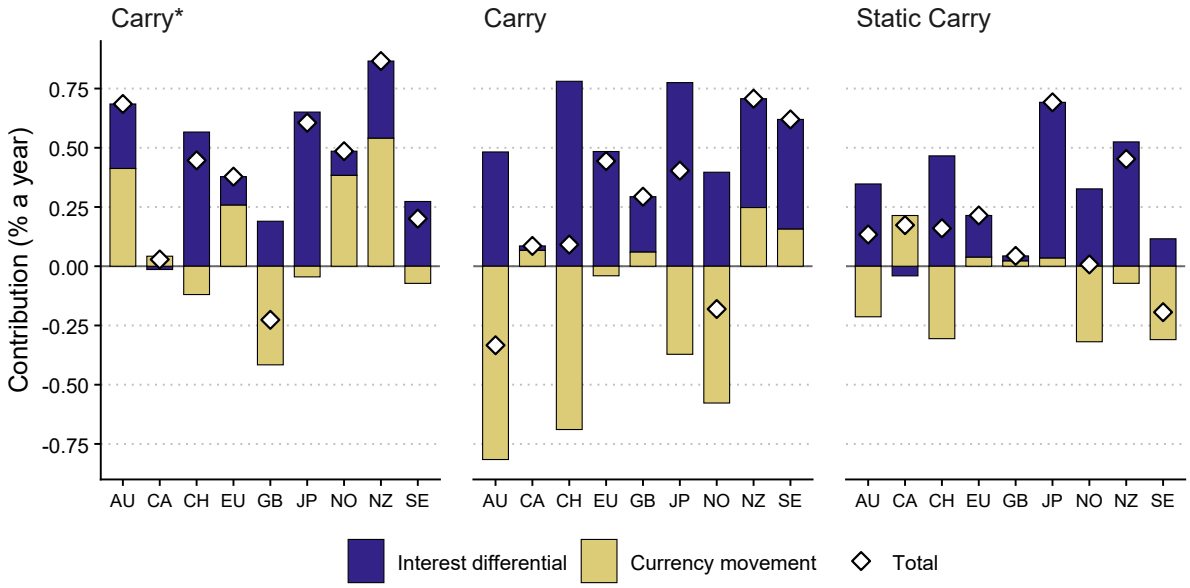


Figure 5: Interest differential and currency movement of the long-short returns by currency. For each currency, bars show its contribution to the mean log excess return of the strategy (its portfolio weight times its interest differential and times its currency movement, averaged over all months). Diamonds mark the total. Contributions sum to the strategy's mean return. EU is the Deutsche mark before January 1999. Sample October 1987 to February 2025, in % a year.

demeans every variable by month, both in the projection and in the regression, so that the slopes use only the differences across currencies within each month, which is the variation a cross-sectional sort exploits. We report Driscoll-Kraay t-statistics of each slope against one. Since the projection uses the full sample, this decomposition is descriptive and not a tradable strategy.

Table 9 shows the results. The slope on the whole forward premium is 0.68 in the pooled regression and 0.69 with month effects (columns 1 and 2), below one but not significantly so in our sample. The split in columns 3 and 4, however, reveals a clear pattern. For the rest of the forward premium, uncovered interest parity holds, with slopes of 1.00 and 0.88 that are statistically indistinguishable from one. For the long-run part, the slopes are -0.64 and -0.07, significantly below one, so currencies with high long-run rates do not depreciate by their long-run differential.[7] In other words, exchange rates offset the part of the interest differential unrelated to the long-run rate, but not the long-run part. This is the pattern implied by the long-run restriction of Feunou et al. (2024), under which a persistent difference in interest rate trends is matched by a persistent currency risk premium rather than by an expected depreciation. This explains why $carry^*$ does not lose through currency movement in columns 2 and 8 of Table 8, while carry, which also sorts on the rest of the differential, does. It also explains why their dynamic trades move in opposite directions. When carry changes positions, it follows movements in the forward premium that exchange rates offset, and when $carry^*$ changes positions, it follows movements in long-run rates that they do not offset. It describes where the return of $carry^*$ comes from, but it does not yet explain why investors earn it.

To understand why the long-run differential is priced, we return to the Fisher equation, $i^* = \pi^* + r^*$, and split the 10-year yield of each country into an inflation part, $\pi^*_{c,t}$, and a real part, $i_{c,t} - \pi^*_{c,t}$. We measure π^* in two ways, both from the real-time CPI data of Subsection 2.1. The first is CPI inflation over the past three years, one of the inputs of the fair value. The second is a trend inflation of the form of Cieslak and Povala (2015), a discounted average of past 12-month inflation over the last 10 years,

$$\tau_{c,t} = \frac{\sum_{j=0}^{119} \nu^j \, \pi_{c,t-j}}{\sum_{j=0}^{119} \nu^j}, \tag{5}$$

where $\pi_{c,t}$ is 12-month CPI inflation. Cieslak and Povala (2015) fix the monthly discount factor at $\nu = 0.987$ for US core inflation. We instead choose it each month, between 0.8

[7] In the Fama regressions of Appendix F, the slope on the forward premium is not significantly different from one with currency or month effects either, and in the split of Hassan and Mano (2019) it is lowest on the currency mean of the forward premium, at 0.30.

Table 9: Uncovered interest parity. The table asks which part of the interest differential the currency movement offsets. Each column regresses each currency's depreciation against the US dollar over the next month, $\Delta s_{c,t+1}$, on its forward premium, the interest differential to the US, pooling the nine currencies (Fama, 1984). Columns 1 and 2 use the whole forward premium. Columns 3 and 4 enter it in two parts: the long-run part, its fitted value from a regression of the forward premium on the $Carry^*$ differential, and the rest. Under uncovered interest parity each slope is one, so the currency depreciates by the interest differential and the strategy earns nothing. A slope below one means the currency loses less than the differential, and a negative slope means it appreciates. In columns 2 and 4 every variable is demeaned each month, so the slopes use only the differences across currencies that a sort uses. The split uses full-sample regressions and is descriptive. Driscoll-Kraay t-statistics of the slope against one in parentheses. October 1987 to February 2025.

	Dependent variable: $\Delta s_{c,t+1}$			
	(1)	(2)	(3)	(4)
Forward premium	0.68	0.69		
	(−1.18)	(−1.52)		
Long-run part			−0.64	−0.07
			(−3.36)	(−2.84)
Rest			1.00	0.88
			(0.01)	(−0.61)
Month effects	No	Yes	No	Yes

and 1, as the value that best forecast inflation over the following three years in the data available at the time. The chosen value is 0.95 or 0.8, so our average gives more weight to recent inflation than theirs.[8] We then sort currencies on the differential to the US of each part, with the same median split as $carry^*$, and compare the resulting strategies with the 10-year yield itself and with $carry^*$.

Panel A of Table 10 reports the Sharpe ratio of each strategy before and after transaction costs, its abnormal return over carry and over carry and static carry, together with the dollar factor, and its risk price in a Fama-MacBeth regression that also includes the dollar factor and carry, on both sets of test assets of Table 4. The results show that the premium is in the inflation part. Currencies sorted on π^* deliver Sharpe ratios of 0.53 and 0.54, close to the 0.57 of $carry^*$, and abnormal returns over carry and static carry of 1.58% and 1.75% per year, with t-statistics of 2.38 and 2.31. Their risk prices are significant on both sets of test assets, and next to them carry is no longer significant at the 5% level. Currencies sorted on the real part, in contrast, deliver Sharpe ratios of only 0.23 and 0.25, abnormal returns that are close to zero and risk prices that are not significant, while carry

[8] Appendix F shows that both measures are close to market measures of long-run inflation, with a cross-sectional correlation of 0.93 with 5-year, 5-year inflation swaps, and that the π^* differential predicts the inflation differential over the next five years.

remains significant next to them on the portfolios of Della Corte and Riddiough (2025). The 10-year yield itself lies in between. It delivers a Sharpe ratio of 0.39 and no significant abnormal return over carry and static carry, and its risk price is significant only on the portfolios of Della Corte and Riddiough (2025). This helps explain why $carry^*$ improves on a sort on the 10-year yield. Since the fair value is a function of inflation and growth only, it leaves out the part of the yield that these variables do not explain.

Panel B of Table 10 shows why this distinction matters. Using market data, we split each nominal 10-year yield differential to the US into the differential in 10-year breakeven inflation and the differential in real yields, measured as the nominal yield minus breakeven inflation. The share of each part is its covariance with the nominal differential divided by the variance of the nominal differential, so the two shares sum to one. Breakeven inflation rates are from LSEG for seven currencies, excluding the Swiss franc and the Norwegian krone and using Germany for the euro, from March 2014 to February 2025. On average across monthly cross-sections, the real part accounts for 70% of the cross-sectional variance of nominal 10-year differentials, and 75% when pooling all currency-months. In other words, most of what makes long-term yields differ across countries is the real part, which earns no premium, while the premium is in the smaller inflation part. A sort on the 10-year yield mixes the two parts, which dilutes the premium.

If the premium is in the inflation part, one may ask why we do not simply sort on π^* and drop the real side of the long-run rate altogether. There are three reasons. First, $carry^*$ does not carry the real part of Table 10 with it. The real part is the yield minus π^*, and it contains everything in the yield that inflation does not explain. The fair value, in contrast, keeps from the real side only what GDP growth explains, which is the component of the real rate that the Fisher equation and growth models tie to the long run, and leaves out the rest of the yield. How much weight growth receives also matters. When a country r^* from 5- or 10-year GDP growth is added one-for-one to trend inflation, as in the decomposition of Bauer and Rudebusch (2020), the Sharpe ratio falls from 0.53 to 0.42,[9] whereas the fair value lets the pooled regression set the weight on growth. Second, a sort on π^* requires choosing a measure of trend inflation, such as the horizon of past inflation or the discount factor in Equation 5. The fair value lets the pooled regression weigh inflation over one and three years, together with growth, in the data available at each date, so no measure of trend inflation is chosen by hand. Third, $carry^*$ performs at

[9] Table F3 in Appendix F reports this decomposition with r^* from 5- or 10-year GDP growth or from the 5-year real short rate, and with π^* measured by trend inflation or 3-year inflation. The Sharpe ratio is between 0.41 and 0.46, and the abnormal return over carry and static carry is not significant in four of the five cases.

least as well as either π^* sort on every measure in Panel A. It delivers a higher Sharpe ratio before and after costs, a larger abnormal return over carry and static carry, 1.93% per year with a t-statistic of 3.16, and a larger risk price on both sets of test assets, while carry has its lowest t-statistic next to $carry^*$. The decomposition therefore serves to locate the premium within the long-run rate, and not to replace the signal.

Taken together, the results in this section indicate that the premium of $carry^*$ is associated with cross-country differences in long-run inflation. $Carry^*$ earns a smaller interest differential than carry but does not lose it through currency movement, and two thirds of its return comes from changing positions as long-run rates move rather than from holding the same currencies throughout the sample. Exchange rates do not offset the long-run part of the interest differential, which is why these changes in positions gain through currency movement while those of carry lose. Within the long-run rate, the premium is in the inflation part rather than in the real part. This evidence is consistent with compensation for long-run inflation risk.[10]

[10] Appendix F adds two results in line with this interpretation. The sorts on π^* gain when US breakeven inflation rises, also after controlling for carry, and exchange rates offset the part of the forward premium explained by the real part but not the part explained by π^*.

Table 10: The inflation part and the real part of the 10-year yield. As in the decomposition $i^* = \pi^* + r^*$ of Bauer and Rudebusch (2020), the 10-year yield is split into an inflation part, π^*, and a real part, the yield minus π^*. π^* is measured in two ways from real-time CPI. The first is 3-year CPI inflation. The second, trend inflation, is a discounted average of past 12-month inflation over 10 years, the form of Cieslak and Povala (2015). They fix the monthly discount factor at 0.987 for US core inflation. Here it is chosen each month, from 0.8 to 1, as the value that best forecast inflation over the next three years in past data. It is 0.95 or 0.8, so the average is shorter than theirs. Panel A sorts currencies on the differential to the US of each part, as $Carry^*$ sorts on its own, with the 10-year yield and $Carry^*$ for comparison. SR is the Sharpe ratio, gross and net of costs. The alphas are over Dollar and Carry, and over Dollar, Carry and Static Carry, in % a year with Newey-West t-statistics (12 lags) in parentheses. The race columns come from a Fama-MacBeth regression with Dollar, Carry and the factor on the test assets of Table 4: our 27 portfolios (own) or the 45 portfolios of Della Corte and Riddiough (2025) (DCR). They give the factor's risk price in % a year with its Shanken t-statistic in parentheses, and the Shanken t-statistic of Carry's risk price. Panel B uses market data to ask how much of the cross-section of nominal 10-year yield differentials to the US is expected inflation and how much is real. Each differential is split into the 10-year breakeven inflation differential and the real yield differential (nominal minus breakeven). The share of each part is its covariance with the nominal differential divided by the variance of the nominal differential, so the two shares sum to one. The first row averages the shares over the monthly cross-sections, and the second pools all currency-months. Observations are months in the first row and currency-months in the second. Breakevens are from LSEG for seven currencies (not CH and NO, and Germany for EU), March 2014 to February 2025. Panel A covers October 1987 to February 2025.

Panel A: Sorting on each part

	SR	SR net	α on Carry (%)	α on Carry and Static Carry (%)	Race, own: factor (t)	Race, own: Carry t	Race, DCR: factor (t)	Race, DCR: Carry t
π^*: trend inflation	0.53	0.51	1.84	1.58	2.40	1.71	3.13	1.95
			(2.35)	(2.38)	(2.32)		(2.88)	
Real part (trend)	0.25	0.22	0.30	−0.06	0.93	1.77	1.25	2.43
			(0.38)	(−0.08)	(0.93)		(1.19)	
π^*: 3-year inflation	0.54	0.52	1.93	1.75	2.36	1.63	3.20	1.75
			(2.27)	(2.31)	(2.40)		(2.92)	
Real part (3-year)	0.23	0.21	0.24	−0.12	0.92	1.74	1.14	2.44
			(0.30)	(−0.17)	(0.91)		(1.07)	
10-year yield	0.39	0.37	0.84	0.38	1.61	2.21	2.57	2.16
			(1.35)	(0.87)	(1.42)		(2.18)	
$Carry^*$	0.57	0.54	2.16	1.93	2.87	1.36	3.78	1.48
			(2.94)	(3.16)	(2.67)		(3.30)	

Panel B: Cross-section of nominal 10-year differentials

	Observations	Breakeven share	Real share
Monthly cross-sections, averaged	132	0.30	0.70
Pooled differentials	912	0.25	0.75

4.2. Bad times, crash risk and oil risk

If the premium of $carry^*$ is compensation for risk, it should be possible to see when the strategy loses money. The carry trade is known to perform poorly in bad times for investors, when global volatility rises and when liquidity dries up (Brunnermeier et al., 2008; Menkhoff et al., 2012a), and its losses are larger when the market falls than its gains when the market rises (Lettau et al., 2014). We therefore look at the returns of $carry^*$ in bad times and at its exposure to measures of crash risk, and compare them with those of carry and static carry.

Panel A of Table 11 reports the mean return of each strategy, annualized, in different states. Recessions are the months that the NBER dates as US recessions. VIX-spike months are the 20% of months with the largest increase in the monthly average of the VIX, which is available from 1990. Carry's worst months are the 10% of months with the lowest return of carry. In each case the state refers to the month in which the return is earned. The table also reports the slope of a regression of each strategy on carry, estimated separately in the months when carry falls and when it rises, and the worst monthly return of each strategy. The results show that $carry^*$ suffers less than the benchmarks in bad times. In US recessions, $carry^*$ earns 1.51% per year, while carry loses 5.86% and static carry loses 2.31%. In VIX-spike months all three strategies lose money, but $carry^*$ loses the least, 6.56% per year against 7.43% for carry and 9.58% for static carry. In the worst 10% of months for carry, $carry^*$ loses 23.14% per year, against 39.95% for carry itself and 25.75% for static carry. Moreover, the exposure of $carry^*$ to carry is symmetric, with a slope of 0.61 when carry falls and 0.62 when it rises, whereas static carry moves with carry much more when carry falls, with slopes of 0.68 and 0.36. The worst month of $carry^*$ is also milder, -7.11%, against -8.89% for carry and -8.31% for static carry. Since recessions and the other states cover a small number of months, these comparisons are descriptive.

Panel B of Table 11 turns to option-implied measures of risk, from LSEG. For currencies, we use the one-month 25-delta risk reversal, a measure of crash risk (Brunnermeier et al., 2008), signed so that a higher value means a more expensive protection against a fall of the foreign currency against the US dollar, and the one-month at-the-money implied volatility. Each month we average each measure across the nine currencies and regress the monthly return of each strategy on the change in this average over the same month, with Newey-West standard errors. The risk reversals start in July 2010, with some currencies entering in 2012 and 2013, and the implied volatilities start in January 1995. For oil, we use the one-month at-the-money implied volatility of WTI crude oil options, the oil skew,

measured as the implied volatility of the 25-delta put minus that of the 25-delta call, and the log change in the front WTI futures price, all from August 2006. The oil regressions also include the dollar factor.

The results show that $carry^*$ is not a safer version of carry when risk is measured from options. All three strategies lose money when FX implied volatility rises, with similar slopes of -0.06 and -0.07 and t-statistics around -9 to -10, in line with Menkhoff et al. (2012a). When crash risk rises, $carry^*$ and static carry lose money, with slopes of -0.21 and -0.26 and t-statistics of -3.86 and -4.46, whereas the slope of carry, -0.08, is not statistically significant. The same pattern appears for oil. $Carry^*$ loses money when oil implied volatility and the oil skew rise, with t-statistics of -3.22 and -2.89, and gains when oil prices rise, with a slope of 0.64 and a t-statistic of 4.31. The corresponding slopes of carry are smaller and not statistically significant for oil volatility and skew, while those of static carry are close to those of $carry^*$.[11] The exposure to oil is consistent with the role of commodity currencies in currency returns (Ready et al., 2017), although the abnormal return of $carry^*$ over the commodity factor in Table 7 shows that this exposure does not account for its premium.

The picture that emerges is that the risk of $carry^*$ has two dimensions. On the one hand, $carry^*$ loses less than carry in recessions, in VIX spikes and in the worst months of carry, and its exposure to carry does not increase when carry falls. On the other hand, $carry^*$ is more exposed than carry to changes in currency crash risk and in oil risk, and as exposed as carry to changes in FX volatility. This exposure to risks that investors care about is consistent with the interpretation of the $carry^*$ premium as compensation for risk, rather than as a free lunch.

5. Robustness

Our results do not depend on the particular fair value used for $carry^*$. Across the 25 versions of the long-run rate listed in Appendix A, every version earns a higher Sharpe ratio than carry and static carry before and after transaction costs, and 20 of the 23 macro versions earn a significant abnormal return over both benchmarks, while neither benchmark earns one over any version (Appendix C). This includes a specification fixed before seeing any returns, which earns an abnormal return of 1.38% per year with a t-statistic of 2.47. In contrast, long-horizon differentials implied by forward exchange rates earn no abnormal return (Appendix D).

[11] Appendix G adds that $carry^*$ has a lower CAPM beta and downside beta than carry and static carry, and that its losses in the largest crash episodes, around the Lehman collapse and during Covid, are similar to or smaller than theirs.

Table 11: Bad times, crash risk and oil risk. Panel A gives mean returns (% a year) in US recessions and expansions, in VIX-spike months (top 20% of VIX changes) and the others, and in Carry's worst 10% of months, the beta to Carry when Carry falls and rises, and the worst month (%). Panel B gives the slope of the monthly return on the change in the mean G10 25-delta risk reversal (from 2010) and one-month at-the-money implied volatility (from 1995), and in WTI implied volatility, skew and return (from August 2006), with Newey-West t-statistics. Option data from LSEG. October 1987 to February 2025.

Panel A: Bad times	Recession / expansion (% a year)	VIX-spike months / other	Carry's worst 10% months	Downside / upside β to Carry	Worst month (%)
Carry*	1.51 / 3.70	−6.56 / 5.69	−23.14	0.61 / 0.62	−7.11
Carry	−5.86 / 2.88	−7.43 / 4.28	−39.95	1.00 / 1.00	−8.89
Static Carry	−2.31 / 2.20	−9.58 / 4.20	−25.75	0.68 / 0.36	−8.31

Panel B: Betas to crash risk and oil risk	Δ risk reversal	Δ FX implied volatility	Δ oil implied volatility	Δ oil skew	Oil return
Carry*	−0.21	−0.06	−0.50	−0.67	0.64
	(−3.86)	(−9.74)	(−3.22)	(−2.89)	(4.31)
Carry	−0.08	−0.06	−0.21	−0.30	0.46
	(−1.59)	(−8.94)	(−1.26)	(−1.07)	(2.98)
Static Carry	−0.26	−0.07	−0.41	−0.70	0.56
	(−4.46)	(−10.32)	(−3.78)	(−3.88)	(3.97)

*Carry** also keeps most of its Sharpe ratio with terciles or rank weights instead of the median split, and its abnormal return over carry and static carry stays significant when any single currency is dropped. Finally, the abnormal return falls from 1.89% per year with the current signal to 0.23% with a six-month-old one, so updating the long-run rate matters (Appendix D).

6. Conclusion

This paper shows that the profitability of currency carry strategies is associated with persistent cross-country differences in long-run interest rates. Motivated by the Fisher equation, we estimate the long-run nominal rate of each country as the real-time fair value of its 10-year government bond yield given inflation and output growth, and sort currencies on its differential to the US. The resulting signal, *carry**, is genuinely distinct from the traditional carry signal. It is far more persistent than the forward premium, ranks currencies differently and turns over less than half as often as carry, and the estimated long-run rates are related to productivity and demographics, the two main drivers of natural interest rates.

A simple strategy that goes long currencies with above-median long-run differentials

and shorts those below delivers an annualized excess return of 3.52%, with a Sharpe ratio of 0.57 before and 0.54 after transaction costs, higher than those of carry and static carry. In Fama-MacBeth regressions on our own portfolios and on the portfolios of Della Corte and Riddiough (2025), $carry^*$ earns a positive and significant risk premium, while carry and static carry lose their significance once $carry^*$ is included. Spanning regressions reveal an asymmetric relationship, in which $carry^*$ earns a significant abnormal return of 1.93% per year after controlling for both benchmarks and the dollar factor, while neither benchmark retains an abnormal return once $carry^*$ is included. The abnormal return of $carry^*$ also survives currency value and momentum, the term spread, global imbalances, country size and commodity factors. Together, these results establish $carry^*$ as a tradable factor that subsumes the pricing information of existing carry benchmarks.

We also show where the premium comes from. Splitting the long-run rate along the Fisher equation, the premium is in its inflation part. Currencies sorted on long-run inflation earn almost the same Sharpe ratio as $carry^*$ and a significant abnormal return over carry and static carry, while currencies sorted on the real part earn neither, even though the real part accounts for most of the cross-country variation in long-term yields in the period covered by breakeven inflation data. This explains why sorting on the long-run rate improves on sorting on the 10-year yield itself. Moreover, $carry^*$ earns a smaller interest differential than carry, but exchange rates do not offset the long-run part of the differential, so that $carry^*$ gains through currency movement where carry loses. This gain comes from changing positions. Unlike carry, whose return is mostly static, two thirds of the return of $carry^*$ comes from updating positions as long-run rates move. $Carry^*$ is therefore not a static ranking of currencies but a middle ground between carry and static carry. Its risk has two dimensions. $Carry^*$ loses less than carry in recessions, in VIX spikes and in the worst months of carry, but it is more exposed than carry to changes in currency crash risk and in oil risk.

Our findings highlight the central role of persistent cross-country differences in long-run interest rates, and in particular in long-run inflation, in explaining carry trade profitability. The evidence is consistent with the premium of $carry^*$ being compensation for long-run inflation risk, rather than a free lunch.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT 5.5 in order to check for grammar errors and improve readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Data Availability Statement

The data supporting the findings of this study are restricted and not publicly available due to legal and commercial reasons. Access to the data may be granted by the corresponding author upon reasonable request, subject to approval and compliance with applicable regulations.

Disclosure of interest

No potential conflict of interest was reported by the author(s).

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Appendix

The appendix contains robustness checks and additional results. Appendix A describes the data and the transaction costs and defines the 24 alternative measures of the long-run rate, which vary the macro features, the horizons, the target of the regression and the estimator. Appendix B shows that the alternatives built from macro data share the persistence and rankings of $carry^*$, and Appendix C repeats the main tests for each of them. Every version earns a higher Sharpe ratio than carry and static carry, and 20 of the 23 macro versions earn a significant abnormal return over both. Appendix D changes the portfolio construction, drops one currency at a time, splits the sample, and sorts on forward-implied differentials and on stale signals. Appendix E controls for value, momentum, the term spread and the global imbalance, size and commodity factors for every version. Appendix F gives additional results on the source of the premium, including uncovered interest parity tests for every version, a comparison of π^* with market measures of long-run inflation and the exposure to inflation news. Appendix G reports exposures to equity market and volatility risk, returns in crash episodes, and exposures to option-implied and oil risk. Appendix H studies a time-series version of $carry^*$ and the predictability of the dollar, and Appendix I relates the long-run rate to the demographics and productivity growth.

A. Data and transaction costs

$Carry^*$ measures the long-run rate with one particular fair value, and several choices go into it: the macro features, the horizons over which they are measured, the target of the regression and the estimator. To show that our results do not depend on these choices, we also build 24 alternative versions of the long-run rate, listed in Table A1. All of them use only data available at the time of portfolio formation, and all of them are sorted in the same way as $carry^*$. They fall into five broad classes. The first changes the macro features of the fair value, using inflation only, longer horizons of GDP growth or a specification fixed before seeing any returns. The second follows the Fisher equation directly and sets the long-run rate to a measure of π^* plus a common r^*, as in Bauer and Rudebusch (2020). The third fits the mean 10-year yield or the mean short-rate differential over the next five years instead of the current 10-year yield, since the long-run rate is in principle the level that rates are expected to reach. The fourth lets the data choose the features each month, by LASSO, elastic net or Bayesian model averaging, which guards against the concern that our specification is the result of a search over features. The fifth uses no macro data at all, with a statistical trend of the 10-year yield from the filter of Hamilton (2018) and the

10-year yield itself. These two show what the macro fair value adds over the information already in long-term yields.

Table A2 describes the data. Panel A reports, for each of the nine currencies, the number of months in the sample, the first month, and the mean forward premium, depreciation and excess return against the US dollar. Panel B reports the mean 10-year yield, the $carry^*$ fair value, inflation and GDP growth of each country. Countries with high yields tend to have high fair values. Japan and Switzerland have the lowest 10-year yields and the lowest fair values, while Australia and New Zealand are among the highest on both.

Table A3 details the transaction costs used for the net returns, constructed as described in Subsection 2.1. Panel A reports the median quoted half-spread of the one-month forward of each currency by decade. Half-spreads are a few basis points and have fallen over time for most currencies, and they are highest for the New Zealand dollar, the Norwegian krone and the Swedish krona. Panel B reports, for each strategy, the Sharpe ratio before and after costs, the annual turnover and the cost per year. The cost per unit of turnover is between 3.4 and 4.0 basis points for every strategy, so differences in costs come from differences in turnover. Carry turns over 10.89 times a year and loses 0.39% per year to costs, against 4.82 times and 0.18% for $carry^*$ and 0.65 times and 0.02% for static carry.

Table A1: Versions of the long-run signal. Every version sorts on $i^*_{c,t} - i^*_{US,t}$ with data available at t. Fair values come from a pooled regression of the 10-year yield on macro features, re-estimated each month on an expanding window. The π^* versions set $i^* = \pi^* + r^*$ (Bauer and Rudebusch, 2020), with trend inflation of the form of Cieslak and Povala (2015) as described in Table 10. The fits to future rates use only forecasts whose outcome is known at t. The spike-and-slab slab is the g-prior of Zellner (1986). Inflation is CPI inflation and GDP growth is real, from OECD real-time vintages from 1999 and current data with publication lags before, with Japanese CPI from Datastream from July 2022.

Version	Definition
Carry*	Fair value of the 10-year yield on 1- and 3-year inflation and 1- and 5-year GDP growth
FV: inflation 1y+3y	Fair value on 1- and 3-year inflation
FV: inflation + GDP 1y, 10y	Fair value on 1- and 3-year inflation and 1- and 10-year GDP growth
FV: fixed in advance	Fair value on 1- and 5-year inflation and 1- and 10-year GDP growth, chosen before seeing returns
π^*: 3y inflation + r^*	π^* is 3-year CPI inflation, with a common r^*
π^*: trend inflation	π^* is a discounted average of past 12-month inflation over 10 years, with the discount chosen each month by how well it forecast inflation, and a common r^*
OLS: future yield	OLS fit of the mean 10-year yield over the next 60 months on 1- and 3-year inflation
OLS: future yield, cross-section	As above, with the yield and the features demeaned by month
OLS: future short rate	As above, for the mean short-rate differential to the US over the next 60 months
LASSO, elastic net 0.5, elastic net CV	Penalised selection among 1-, 3- and 5-year inflation and 1-, 5- and 10-year GDP growth, for the same three targets. Each month, the penalty is chosen by 5-fold cross-validation on the data available at that date, with folds that are contiguous blocks of months, taking the largest penalty within one standard error of the minimum error. Elastic net CV also chooses, with the same folds, the weight on the LASSO penalty from 0 (ridge) to 1 (LASSO) in steps of 0.25
LASSO / EN: BR structure	Trend inflation plus an r^* selected from GDP growth differentials (the penalty sets r^* to zero), with the cross-validation above
Spike-and-slab	Bayesian model averaging over the same features, for the three targets and the BR structure. Each coefficient is zero (spike) or follows Zellner's g-prior with g equal to the number of observations (slab), with flat priors on the intercept and the error variance and equal prior weight on every feature set (inclusion probability 0.5). All sets are enumerated, and i^* is the model-averaged posterior mean
Hamilton	24-month rolling Hamilton filter of the 10-year yield (12 lags)
10-year yield	The 10-year yield itself

Table A2: Summary statistics of the data. Panel A: the nine currencies against the dollar, with the mean log forward premium, depreciation and excess return and the excess return's standard deviation, annualized in %. Panel B: means of the 10-year yield, the $Carry^*$ fair value, real-time 12-month inflation and 1-year GDP growth, and the yield's standard deviation, in %. EU is Germany before January 1999. FX data from Datastream (LSEG), macro data from the OECD. October 1987 to February 2025.

Panel A: Currencies						
	Months	First month	Forward premium	Depreciation	Excess return	Excess return s.d.
AU	435	Dec 1988	1.43	0.88	0.55	11.11
CA	449	Oct 1987	0.27	0.24	0.03	7.53
CH	449	Oct 1987	−1.80	−1.27	−0.53	10.44
EU	449	Oct 1987	−0.66	0.12	−0.78	9.89
GB	449	Oct 1987	0.97	0.77	0.19	9.41
JP	449	Oct 1987	−2.58	0.22	−2.80	10.55
NO	449	Oct 1987	1.25	1.27	−0.03	11.18
NZ	435	Dec 1988	2.17	0.30	1.87	11.44
SE	434	Jan 1989	0.48	1.28	−0.80	11.17
Panel B: Macro data						
	10-year yield	10-year yield s.d.	Carry* fair value	12-month inflation	1-year GDP growth	
AU	5.61	2.99	6.21	2.89	2.73	
CA	4.71	2.79	5.71	2.33	2.05	
CH	2.42	2.04	4.74	1.18	1.45	
EU	3.71	2.60	5.44	2.22	1.49	
GB	4.82	3.00	6.16	3.08	1.65	
JP	1.90	1.88	4.13	0.70	1.30	
NO	5.01	3.11	5.93	2.62	1.99	
NZ	5.76	2.65	6.05	2.50	2.31	
SE	4.46	3.57	5.76	2.39	1.90	
US	4.49	2.18	6.09	2.76	2.55	

Table A3: Transaction costs from quoted bid-ask spreads. Panel A: the half-spread (ask − bid) / (ask + bid) of the one-month forward, monthly median of daily quotes, in basis points, median by decade (Datastream, from May 1990). The Deutsche mark (DE) takes its LSEG spot half-spread (from October 1989) plus the euro's forward-minus-spot gap, and months before the first quote take the median of the first 36 quoted months. Panel B: Sharpe ratios gross and net, annual two-way turnover, cost in % a year and cost per unit of turnover in basis points. Quoted spreads exceed traded spreads. Versions as in Table A1.

Panel A: Half-spreads of the one-month forward (bp)

	1980s	1990s	2000s	2010s	2020s	Months: quoted / DE spot + gap / filled
AU	4.37	4.11	3.55	2.42	1.66	418 / 0 / 17
CA	2.60	2.48	2.48	1.81	1.58	418 / 0 / 31
CH	4.18	3.91	2.28	2.43	2.42	418 / 0 / 31
DE	3.37	2.82				0 / 110 / 24
EU		2.54	1.88	1.36	1.63	315 / 0 / 0
GB	3.49	3.24	1.59	1.37	1.82	418 / 0 / 31
JP	3.62	3.31	2.38	1.54	1.28	418 / 0 / 31
NO	4.97	3.37	4.52	3.59	4.96	418 / 0 / 31
NZ	7.42	7.15	7.74	3.66	3.11	418 / 0 / 17
SE	2.29	6.11	5.31	2.64	5.73	418 / 0 / 16

Panel B: Costs by strategy

	SR gross	SR net	Turnover (a year)	Cost (% a year)	Cost per unit of turnover (bp)
Hamilton	0.47	0.44	4.88	0.17	3.55
10-year yield	0.39	0.37	2.56	0.09	3.66
FV: inflation 1y+3y	0.62	0.60	3.29	0.12	3.67
Carry*	0.57	0.54	4.82	0.18	3.70
FV: inflation + GDP 1y, 10y	0.60	0.56	5.22	0.19	3.57
FV: fixed in advance	0.50	0.47	4.66	0.18	3.82
π^*: 3y inflation + r^*	0.54	0.52	2.86	0.11	3.73
π^*: trend inflation	0.53	0.51	2.09	0.08	3.72
OLS: future yield	0.61	0.59	3.28	0.12	3.59
OLS: future yield, cross-section	0.59	0.57	3.55	0.13	3.55
OLS: future short rate	0.59	0.57	3.93	0.14	3.66
LASSO: future yield	0.46	0.44	3.16	0.11	3.45
LASSO: future yield, cross-section	0.47	0.45	2.72	0.11	4.00
LASSO: future short rate	0.46	0.43	3.59	0.14	3.96
Elastic net 0.5: future yield	0.49	0.47	3.69	0.14	3.88
Elastic net 0.5: future yield, cross-section	0.42	0.40	2.92	0.11	3.91
Elastic net 0.5: future short rate	0.51	0.49	3.59	0.14	3.92
Elastic net CV: future yield	0.49	0.47	3.67	0.14	3.88
Elastic net CV: future yield, cross-section	0.51	0.49	2.83	0.11	3.90
Elastic net CV: future short rate	0.52	0.50	3.91	0.15	3.75
LASSO / EN: BR structure	0.55	0.54	2.09	0.08	3.71
Spike-and-slab: future yield	0.41	0.39	4.16	0.14	3.38
Spike-and-slab: future yield, cross-section	0.48	0.46	4.10	0.14	3.45
Spike-and-slab: future short rate	0.49	0.47	4.10	0.14	3.50
Spike-and-slab: BR structure	0.51	0.49	2.40	0.09	3.77
Carry	0.35	0.29	10.89	0.39	3.55
Static Carry	0.30	0.30	0.65	0.02	3.66

B. Alternative measures of the long-run rate

Table B1 extends Table 1 to the 25 versions of the long-run rate described in Appendix A. All the versions built from macro data share the properties of $carry^*$. They are far more persistent than the forward premium, with a first-order autocorrelation between 0.90 and 0.99 against 0.53, they rank currencies in a more stable way, with a rank stability between 0.93 and 0.98 against 0.68, and their correlation with the forward premium is below one half. Figure B1 shows that the macro versions in the figure rank currencies much like $carry^*$, with average rank correlations between 0.82 and 0.86, and quite differently from carry and static carry, with rank correlations between 0.47 and 0.55. The Hamilton trend and the 10-year yield stand apart, as their rankings are closer to that of static carry, with rank correlations of 0.80 and 0.82.

Table B2 shows the same pattern in returns. The return of $carry^*$ has a correlation between 0.76 and 0.84 with those of the other macro versions in the table, of 0.62 with carry, 0.60 with static carry and 0.20 with the dollar factor, while the returns of the Hamilton trend and the 10-year yield have correlations of 0.80 and 0.81 with static carry. Figure B2 shows the positions behind these returns. $Carry^*$ holds the Japanese yen short almost throughout the sample and the Swiss franc short since the late 1990s, but the other currencies move between the long and short portfolios as their long-run rates change, as with the Australian dollar, held long for most of the 2000s and 2010s. Carry switches positions much more often, while static carry keeps the same positions after its first months.

Table B1: Summary statistics of every signal. Differentials to the US in percentage points. TS s.d. is the mean time-series standard deviation across currencies and XS s.d. the mean cross-sectional one across months. AC(k) is the mean k-month autocorrelation. Corr. with fp is the pooled correlation with the forward premium and XS rank corr. the mean monthly Spearman correlation. Rank stability and same side as Carry as in Table 1. Versions as in Table A1. October 1987 to February 2025.

Signal	Mean	TS s.d.	XS s.d.	AC(1)	AC(12)	Rank stability	Corr. with fp	XS rank corr. with fp	Same side as Carry
Hamilton	−0.18	1.15	1.56	0.88	0.67	0.94	0.60	0.69	0.68
10-year yield	−0.18	1.09	1.53	0.98	0.76	0.98	0.61	0.71	0.70
FV: inflation 1y+3y	−0.43	0.76	0.88	0.98	0.66	0.97	0.45	0.55	0.60
Carry*	−0.51	0.83	0.95	0.91	0.50	0.94	0.45	0.55	0.62
FV: inflation + GDP 1y, 10y	−0.44	0.93	1.00	0.90	0.44	0.93	0.45	0.51	0.60
FV: fixed in advance	−0.42	0.83	0.98	0.93	0.53	0.94	0.47	0.54	0.62
π^*: 3y inflation + r^*	−0.50	1.02	1.17	0.98	0.72	0.97	0.47	0.55	0.60
π^*: trend inflation	−0.44	1.03	1.17	0.99	0.60	0.98	0.47	0.54	0.62
OLS: future yield	−0.36	0.66	0.77	0.98	0.64	0.97	0.44	0.55	0.60
OLS: future yield, cross-section	−0.35	0.69	0.80	0.98	0.62	0.97	0.47	0.55	0.61
OLS: future short rate	−0.46	0.91	1.03	0.97	0.51	0.96	0.45	0.55	0.61
LASSO: future yield	−0.39	0.56	0.78	0.97	0.69	0.97	0.49	0.59	0.65
LASSO: future yield, cross-section	−0.24	0.45	0.55	0.98	0.73	0.98	0.47	0.58	0.63
LASSO: future short rate	−0.34	0.56	0.69	0.96	0.48	0.97	0.45	0.57	0.63
Elastic net 0.5: future yield	−0.40	0.58	0.79	0.95	0.60	0.96	0.48	0.59	0.65
Elastic net 0.5: future yield, cross-section	−0.24	0.45	0.54	0.98	0.70	0.98	0.48	0.57	0.63
Elastic net 0.5: future short rate	−0.32	0.55	0.67	0.97	0.55	0.97	0.46	0.57	0.62
Elastic net CV: future yield	−0.40	0.55	0.76	0.95	0.56	0.96	0.48	0.60	0.65
Elastic net CV: future yield, cross-section	−0.25	0.43	0.53	0.97	0.65	0.98	0.46	0.58	0.63
Elastic net CV: future short rate	−0.39	0.57	0.70	0.95	0.50	0.96	0.46	0.59	0.65
LASSO / EN: BR structure	−0.44	1.04	1.17	0.99	0.61	0.98	0.48	0.54	0.62
Spike-and-slab: future yield	−0.52	0.84	0.99	0.93	0.56	0.95	0.48	0.55	0.63
Spike-and-slab: future yield, cross-section	−0.48	0.76	0.90	0.91	0.43	0.95	0.49	0.57	0.64
Spike-and-slab: future short rate	−0.65	0.96	1.14	0.92	0.41	0.95	0.47	0.57	0.64
Spike-and-slab: BR structure	−0.46	1.09	1.17	0.99	0.59	0.98	0.47	0.52	0.62
Carry	0.16	3.38	2.54	0.53	0.40	0.68	1.00	1.00	1.00
Static Carry	2.54	0.00	5.41			1.00	0.36	0.60	0.62

Table B2: Return correlations. Correlations of the monthly returns of the long-short strategies and Dollar. Versions as in Table A1. October 1987 to February 2025.

	Carry*	FV: inflation 1y+3y	π^*: 3y inflation + r^*	π^*: trend inflation	OLS: future yield	Hamilton	10-year yield	Carry	Static Carry	Dollar
Carry*	1.00	0.83	0.80	0.76	0.84	0.71	0.70	0.62	0.60	0.20
FV: inflation 1y+3y	0.83	1.00	0.96	0.84	0.99	0.65	0.61	0.55	0.52	0.15
π^*: 3y inflation + r^*	0.80	0.96	1.00	0.84	0.96	0.64	0.61	0.57	0.54	0.17
π^*: trend inflation	0.76	0.84	0.84	1.00	0.84	0.72	0.69	0.59	0.62	0.14
OLS: future yield	0.84	0.99	0.96	0.84	1.00	0.65	0.61	0.55	0.53	0.15
Hamilton	0.71	0.65	0.64	0.72	0.65	1.00	0.95	0.71	0.80	0.14
10-year yield	0.70	0.61	0.61	0.69	0.61	0.95	1.00	0.71	0.81	0.12
Carry	0.62	0.55	0.57	0.59	0.55	0.71	0.71	1.00	0.61	0.16
Static Carry	0.60	0.52	0.54	0.62	0.53	0.80	0.81	0.61	1.00	0.37
Dollar	0.20	0.15	0.17	0.14	0.15	0.14	0.12	0.16	0.37	1.00

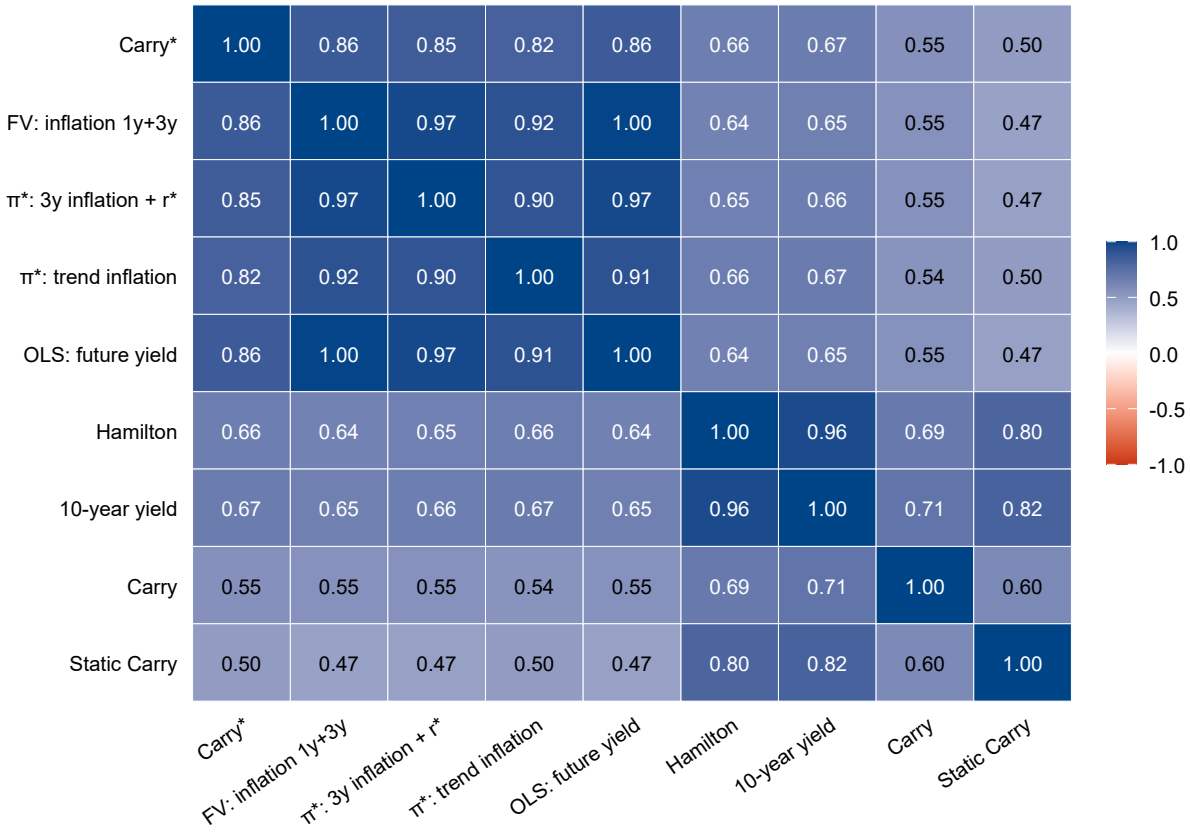


Figure B1: Average cross-sectional rank correlations of the sorting signals. Each cell is the Spearman rank correlation across currencies between two signals, averaged over the months from October 1987 to February 2025. The long-run signals are differentials to the US. Carry is the forward premium and static carry its mean over each currency's first four years.

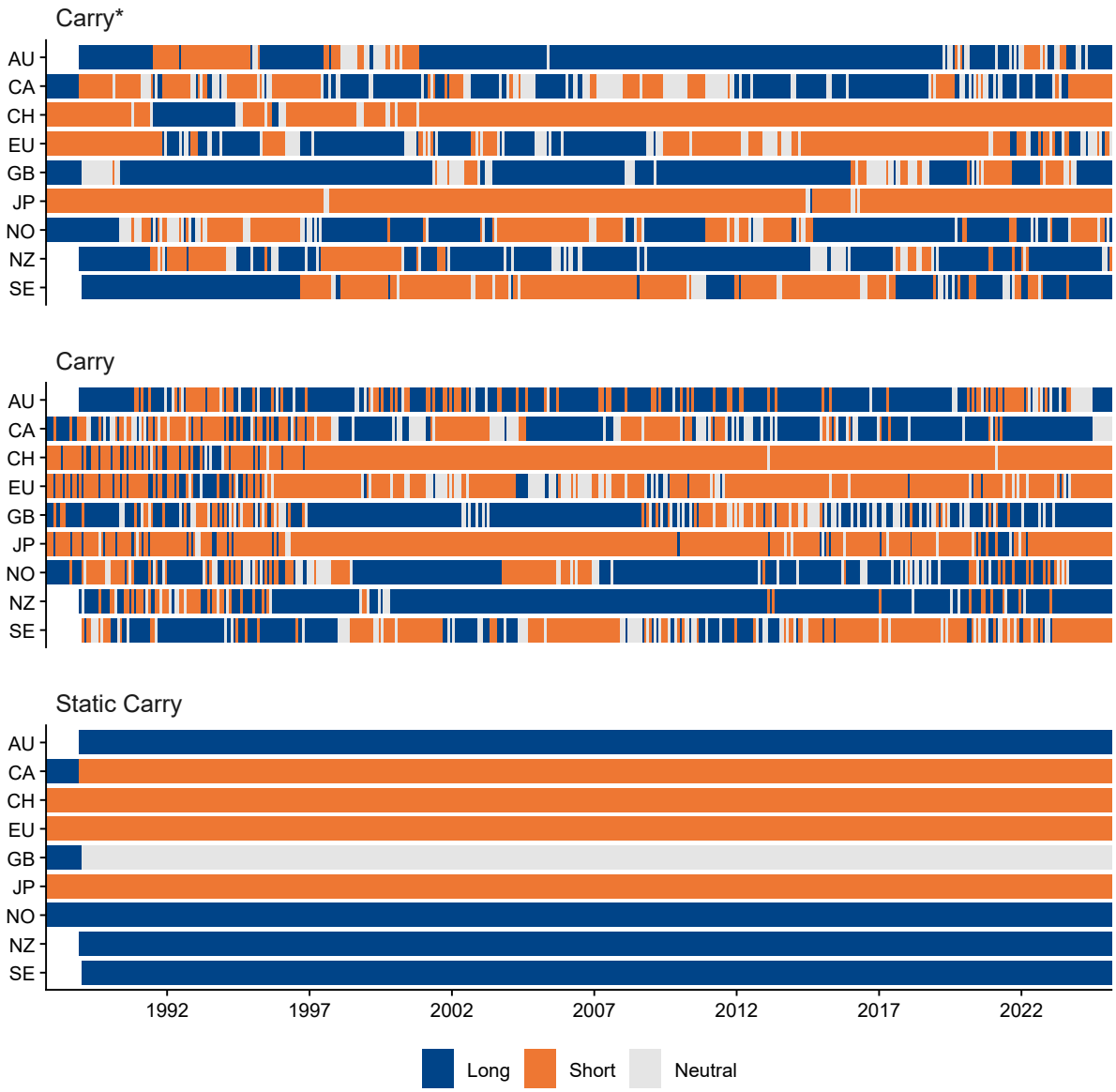


Figure B2: Positions by currency and month. Each row shows whether the currency is in the long (high) or short (low) portfolio of the *carry**, carry and static carry strategies. With an odd number of currencies the median currency is neutral (not held). EU is the Deutsche mark before January 1999. Sample October 1987 to February 2025.

C. Performance and pricing of the alternative measures

We repeat the main tests of Section 3 for each of the 25 versions of the long-run rate. Table C1 reports the performance of each version as in Table 3, Table C2 its price of risk as in Table 4, and Table C3 its abnormal return over carry and static carry and theirs over it.

The results do not depend on the particular version. Every version earns a higher Sharpe ratio than carry and static carry, before and after transaction costs. The price of risk of the version is significant at 5% on the portfolios of Della Corte and Riddiough (2025) for all 25 versions, both alone and next to carry, and on our own portfolios for 23 versions alone and 19 next to carry, while carry is significant next to the version in at most 5 of 25 cases. Of the 23 versions built from macro data, 20 earn a significant abnormal return over carry and static carry, while neither benchmark earns a significant abnormal return over any version. Two patterns stand out. First, the simple pooled OLS fair values generally perform better than the versions that select features by penalised regression or Bayesian model averaging, and the version fixed before seeing any returns also earns a significant abnormal return over both benchmarks, so the premium does not come from a careful choice of features. Second, the Hamilton trend and the 10-year yield, which keep the part of long-term yields that inflation and growth do not explain, earn Sharpe ratios of 0.47 and 0.39, below every pooled OLS fair value, and no significant abnormal return over both benchmarks. This is in line with Subsection 4.1, where the premium is in the part of the long-run rate explained by inflation.

Table C1: Performance of every signal. Statistics as in Table 3. Versions as in Table A1. October 1987 to February 2025.

Signal	Mean (%)	Std. Dev. (%)	Sharpe ratio	t	Turnover	Sharpe (net)	Sortino	Omega	Skewness	Kurtosis	Cum. return	Max DD
Hamilton	2.97	6.36	0.47	2.82	4.88	0.44	0.67	1.42	−0.36	0.95	1.81	−0.22
10-year yield	2.55	6.58	0.39	2.35	2.56	0.37	0.52	1.34	−0.59	1.21	1.39	−0.26
FV: inflation 1y+3y	3.55	5.76	0.62	3.65	3.29	0.60	0.82	1.59	−0.59	1.40	2.54	−0.22
Carry*	3.52	6.16	0.57	3.45	4.82	0.54	0.82	1.53	−0.43	1.19	2.47	−0.23
FV: inflation + GDP 1y, 10y	3.58	6.01	0.60	3.57	5.22	0.56	0.81	1.57	−0.53	1.76	2.56	−0.21
FV: fixed in advance	2.95	5.88	0.50	3.02	4.66	0.47	0.69	1.46	−0.54	1.62	1.82	−0.17
π^*: 3y inflation + r^*	3.09	5.69	0.54	3.24	2.86	0.52	0.73	1.51	−0.61	1.52	1.98	−0.24
π^*: trend inflation	3.12	5.93	0.53	3.16	2.09	0.51	0.70	1.49	−0.62	1.13	2.01	−0.24
OLS: future yield	3.51	5.73	0.61	3.64	3.28	0.59	0.81	1.59	−0.63	1.48	2.49	−0.22
OLS: future yield, cross-section	3.40	5.72	0.59	3.53	3.55	0.57	0.78	1.57	−0.66	1.53	2.35	−0.22
OLS: future short rate	3.54	5.97	0.59	3.48	3.93	0.57	0.79	1.57	−0.57	1.32	2.52	−0.25
LASSO: future yield	2.74	5.91	0.46	2.87	3.16	0.44	0.69	1.41	−0.31	0.93	1.61	−0.20
LASSO: future yield, cross-section	2.69	5.74	0.47	2.86	2.72	0.45	0.66	1.42	−0.48	0.78	1.57	−0.21
LASSO: future short rate	2.62	5.72	0.46	2.73	3.59	0.43	0.63	1.41	−0.57	1.17	1.50	−0.23
Elastic net 0.5: future yield	2.91	5.90	0.49	3.06	3.69	0.47	0.72	1.44	−0.38	1.21	1.78	−0.16
Elastic net 0.5: future yield, cross-section	2.42	5.71	0.42	2.55	2.92	0.40	0.60	1.37	−0.50	0.89	1.33	−0.21
Elastic net 0.5: future short rate	2.91	5.71	0.51	2.99	3.59	0.49	0.70	1.47	−0.54	1.21	1.79	−0.23
Elastic net CV: future yield	2.94	5.97	0.49	3.07	3.67	0.47	0.72	1.45	−0.37	1.16	1.81	−0.16
Elastic net CV: future yield, cross-section	3.02	5.93	0.51	3.11	2.83	0.49	0.69	1.47	−0.61	1.17	1.89	−0.21
Elastic net CV: future short rate	3.20	6.16	0.52	3.10	3.91	0.50	0.71	1.47	−0.61	1.25	2.08	−0.22
LASSO / EN: BR structure	3.30	5.96	0.55	3.31	2.09	0.54	0.74	1.52	−0.60	1.11	2.21	−0.24
Spike-and-slab: future yield	2.45	5.95	0.41	2.58	4.16	0.39	0.63	1.35	−0.28	0.36	1.34	−0.17
Spike-and-slab: future yield, cross-section	2.91	6.01	0.48	2.98	4.10	0.46	0.70	1.43	−0.46	0.86	1.78	−0.18
Spike-and-slab: future short rate	3.01	6.14	0.49	2.97	4.10	0.47	0.69	1.44	−0.50	1.09	1.87	−0.23
Spike-and-slab: BR structure	3.19	6.26	0.51	3.01	2.40	0.49	0.66	1.47	−0.67	1.35	2.06	−0.27
Carry	2.18	6.21	0.35	2.18	10.89	0.29	0.47	1.31	−0.64	1.89	1.10	−0.18
Static Carry	1.84	6.04	0.30	1.84	0.65	0.30	0.44	1.25	−0.41	1.38	0.86	−0.24

Table C2: Fama-MacBeth risk prices of every version. Risk price of the version next to Dollar, and in the race with Dollar and Carry with Carry's risk price, as in columns (1) and (4) of Table 4, in % a year with Shanken (1992) standard errors in parentheses. Own test assets: 27 rank portfolios of the nine currencies on the version, the forward premium and Static Carry (from January 1989). DCR test assets: the 45 G10 quintile portfolios of Della Corte and Riddiough (2025). The last row counts versions significant at 5%. Versions as in Table A1.

Version	Own 27: version next to Dollar	Own 27, race: version	Own 27, race: Carry	DCR 45: version next to Dollar	DCR 45, race: version	DCR 45, race: Carry
Hamilton	2.18 (1.12)	2.01 (1.11)	1.95 (1.08)	3.07 (1.15)	2.92 (1.14)	2.30 (1.16)
10-year yield	1.99 (1.15)	1.61 (1.14)	2.40 (1.09)	2.90 (1.18)	2.57 (1.18)	2.55 (1.18)
FV: inflation 1y+3y	2.87 (1.11)	2.86 (1.00)	1.66 (1.13)	3.37 (1.12)	3.46 (1.10)	1.86 (1.11)
Carry*	2.74 (1.15)	2.87 (1.07)	1.51 (1.11)	3.47 (1.15)	3.78 (1.14)	1.64 (1.11)
FV: inflation + GDP 1y, 10y	2.76 (1.14)	2.81 (1.06)	1.59 (1.13)	3.37 (1.14)	3.51 (1.12)	1.83 (1.10)
FV: fixed in advance	2.42 (1.10)	2.35 (1.04)	1.66 (1.09)	3.15 (1.13)	3.12 (1.07)	2.04 (1.10)
π^*: 3y inflation + r^*	2.56 (1.09)	2.36 (0.98)	1.83 (1.12)	3.20 (1.11)	3.20 (1.10)	1.95 (1.11)
π^*: trend inflation	2.59 (1.11)	2.40 (1.03)	1.87 (1.10)	3.25 (1.15)	3.13 (1.09)	2.14 (1.10)
OLS: future yield	2.80 (1.11)	2.84 (0.99)	1.55 (1.13)	3.36 (1.11)	3.45 (1.10)	1.84 (1.11)
OLS: future yield, cross-section	2.77 (1.10)	2.80 (0.99)	1.58 (1.13)	3.33 (1.11)	3.43 (1.09)	1.85 (1.11)
OLS: future short rate	2.86 (1.13)	2.86 (1.03)	1.64 (1.13)	3.40 (1.14)	3.44 (1.12)	1.94 (1.11)
LASSO: future yield	2.21 (1.08)	2.00 (1.05)	1.88 (1.09)	3.10 (1.13)	2.92 (1.09)	2.26 (1.15)
LASSO: future yield, cross-section	2.25 (1.08)	1.98 (1.01)	1.82 (1.10)	3.10 (1.13)	2.89 (1.05)	2.19 (1.10)
LASSO: future short rate	2.16 (1.08)	1.96 (1.00)	1.76 (1.11)	2.93 (1.10)	2.86 (1.08)	2.06 (1.10)
Elastic net 0.5: future yield	2.32 (1.08)	2.21 (1.04)	1.80 (1.08)	3.13 (1.13)	3.07 (1.08)	2.14 (1.12)
Elastic net 0.5: future yield, cross-section	2.16 (1.08)	1.81 (1.00)	1.94 (1.10)	2.96 (1.12)	2.78 (1.06)	2.15 (1.09)
Elastic net 0.5: future short rate	2.30 (1.08)	2.19 (1.00)	1.69 (1.12)	3.01 (1.09)	2.97 (1.08)	2.06 (1.09)
Elastic net CV: future yield	2.38 (1.09)	2.24 (1.05)	1.86 (1.08)	3.10 (1.13)	2.99 (1.09)	2.19 (1.12)
Elastic net CV: future yield, cross-section	2.44 (1.11)	2.32 (1.03)	1.75 (1.11)	3.04 (1.12)	3.00 (1.12)	2.03 (1.10)
Elastic net CV: future short rate	2.54 (1.14)	2.54 (1.07)	1.63 (1.11)	3.16 (1.15)	3.18 (1.15)	2.00 (1.08)
LASSO / EN: BR structure	2.65 (1.11)	2.48 (1.04)	1.86 (1.09)	3.30 (1.15)	3.22 (1.10)	2.12 (1.10)
Spike-and-slab: future yield	2.21 (1.11)	1.98 (1.03)	1.84 (1.11)	3.06 (1.15)	3.08 (1.14)	2.00 (1.13)
Spike-and-slab: future yield, cross-section	2.39 (1.11)	2.31 (1.06)	1.76 (1.09)	3.07 (1.14)	3.10 (1.12)	2.01 (1.07)
Spike-and-slab: future short rate	2.48 (1.13)	2.28 (1.07)	1.88 (1.09)	3.11 (1.15)	3.09 (1.13)	2.01 (1.08)
Spike-and-slab: BR structure	2.69 (1.15)	2.52 (1.08)	1.85 (1.10)	3.31 (1.19)	3.24 (1.17)	2.06 (1.11)
Significant at 5%	23 of 25	19 of 25	1 of 25	25 of 25	25 of 25	5 of 25

Table C3: Spanning alphas of every version. Alpha of the version over Carry, Static Carry and both, and of Carry and Static Carry over the version, with Dollar in every regression, in % a year. Newey-West t-statistics (12 lags) in parentheses. Versions as in Table A1. October 1987 to February 2025.

Version	α on Carry	α on Static Carry	α on both	Carry's α on version	Static Carry's α on version	Carry's α on version + Static Carry
Hamilton	1.34 (2.11)	1.21 (2.10)	0.91 (1.83)	0.10 (0.19)	−0.31 (−0.60)	0.13 (0.25)
10-year yield	0.84 (1.35)	0.69 (1.24)	0.38 (0.87)	0.45 (0.85)	0.05 (0.11)	0.45 (0.84)
FV: inflation 1y+3y	2.39 (3.32)	2.51 (3.83)	2.21 (3.50)	0.02 (0.03)	0.02 (0.03)	0.01 (0.02)
Carry*	2.16 (2.94)	2.29 (3.60)	1.93 (3.16)	−0.09 (−0.14)	−0.10 (−0.19)	−0.05 (−0.09)
FV: inflation + GDP 1y, 10y	2.30 (3.03)	2.49 (3.52)	2.13 (3.12)	−0.09 (−0.13)	0.01 (0.02)	−0.09 (−0.16)
FV: fixed in advance	1.68 (2.46)	1.64 (2.85)	1.38 (2.47)	0.24 (0.36)	−0.07 (−0.13)	0.26 (0.45)
π^*: 3y inflation + r^*	1.93 (2.27)	2.05 (2.79)	1.75 (2.31)	0.23 (0.33)	0.19 (0.33)	0.13 (0.22)
π^*: trend inflation	1.84 (2.35)	1.86 (2.89)	1.58 (2.38)	0.20 (0.29)	−0.02 (−0.03)	0.21 (0.32)
OLS: future yield	2.37 (3.29)	2.47 (3.85)	2.18 (3.50)	0.05 (0.07)	0.01 (0.01)	0.04 (0.08)
OLS: future yield, cross-section	2.23 (2.94)	2.34 (3.57)	2.04 (3.12)	0.07 (0.11)	0.02 (0.04)	0.06 (0.11)
OLS: future short rate	2.30 (3.01)	2.47 (3.53)	2.12 (3.13)	0.02 (0.03)	0.08 (0.13)	−0.02 (−0.03)
LASSO: future yield	1.38 (1.93)	1.31 (2.26)	1.05 (1.82)	0.29 (0.41)	−0.05 (−0.10)	0.31 (0.46)
LASSO: future yield, cross-section	1.48 (2.06)	1.43 (2.39)	1.19 (1.98)	0.45 (0.61)	0.12 (0.21)	0.40 (0.61)
LASSO: future short rate	1.36 (2.02)	1.41 (2.49)	1.11 (2.01)	0.36 (0.55)	0.19 (0.33)	0.29 (0.50)
Elastic net 0.5: future yield	1.54 (2.42)	1.51 (2.79)	1.22 (2.42)	0.14 (0.23)	−0.14 (−0.28)	0.19 (0.32)
Elastic net 0.5: future yield, cross-section	1.19 (1.63)	1.19 (1.98)	0.92 (1.50)	0.55 (0.78)	0.30 (0.53)	0.43 (0.67)
Elastic net 0.5: future short rate	1.63 (2.34)	1.71 (2.69)	1.39 (2.30)	0.17 (0.24)	0.05 (0.08)	0.15 (0.26)
Elastic net CV: future yield	1.55 (2.41)	1.50 (2.82)	1.21 (2.39)	0.13 (0.20)	−0.18 (−0.35)	0.19 (0.31)
Elastic net CV: future yield, cross-section	1.71 (2.45)	1.81 (2.91)	1.48 (2.43)	0.17 (0.26)	0.09 (0.14)	0.13 (0.23)
Elastic net CV: future short rate	1.76 (2.55)	1.90 (2.85)	1.53 (2.55)	0.06 (0.10)	0.02 (0.04)	0.05 (0.09)
LASSO / EN: BR structure	2.00 (2.52)	2.01 (3.07)	1.72 (2.56)	0.09 (0.12)	−0.13 (−0.22)	0.14 (0.21)
Spike-and-slab: future yield	1.11 (1.58)	1.15 (1.81)	0.84 (1.36)	0.48 (0.67)	0.27 (0.44)	0.38 (0.60)
Spike-and-slab: future yield, cross-section	1.50 (2.15)	1.56 (2.56)	1.22 (2.11)	0.14 (0.22)	−0.02 (−0.04)	0.15 (0.25)
Spike-and-slab: future short rate	1.61 (2.23)	1.65 (2.51)	1.32 (2.21)	0.21 (0.35)	0.03 (0.05)	0.20 (0.38)
Spike-and-slab: BR structure	1.82 (2.18)	1.88 (2.70)	1.56 (2.16)	0.24 (0.36)	0.08 (0.14)	0.20 (0.34)

D. Portfolio construction and signal timing

Table D1 replaces the median split with terciles or with rank weights. *Carry** keeps most of its Sharpe ratio, 0.52 with terciles and 0.49 with rank weights against 0.57 with the median split, and remains above carry and static carry, whose Sharpe ratios are at most 0.41 under any scheme. Its abnormal return over carry and static carry stays positive, 1.73% per year with terciles (t-statistic 2.28) and 1.39% with rank weights (t-statistic 1.78). Table D2 drops one currency at a time. The Sharpe ratio of *carry** stays between 0.45 and 0.65, it beats carry and static carry in all nine samples, and its abnormal return over both is significant at 5% in all nine. Table D3 extends the subperiod analysis of Table 6 to the main versions. The macro versions keep Sharpe ratios between 0.30 and 0.42 after 2006, when carry falls to 0.18, while the Hamilton trend and the 10-year yield earn little after 2006 and no abnormal return.

Two further tables concern the signal itself. Table D4 sorts on long-horizon interest differentials implied by forward exchange rates, available from November 1996. These market-based differentials earn Sharpe ratios between 0.32 and 0.34 and no significant abnormal return over carry and static carry, while *carry** earns 0.44 and an abnormal

return of 1.20% per year with a t-statistic of 2.47 over the same period. What matters is therefore the long-run rate implied by inflation and growth, not the long-horizon differential priced in forward markets. Table D5 sorts on signals between one and 24 months old. The abnormal return of $carry^*$ falls from 1.89% per year with the current signal to 0.23% with a six-month-old one, even though the signal itself is persistent. This is in line with the large dynamic part of its return in Subsection 4.1, since the premium is earned by updating positions as long-run rates move.

Table D1: Portfolio construction. Median split, terciles and rank weights: Sharpe ratio gross and net, and alpha over Dollar, Carry and Static Carry in % a year (Newey-West t-statistic). Panel B gives Carry and Static Carry. Versions as in Table A1. October 1987 to February 2025.

Panel A: Versions

	Median (paper): SR / net	Median (paper): α (%)	Terciles: SR / net	Terciles: α (%)	Rank weights: SR / net	Rank weights: α (%)
Hamilton	0.47 / 0.44	0.91 (1.83)	0.40 / 0.37	0.46 (0.83)	0.41 / 0.39	0.46 (0.92)
10-year yield	0.39 / 0.37	0.38 (0.87)	0.38 / 0.36	0.22 (0.45)	0.39 / 0.38	0.30 (0.58)
FV: inflation 1y+3y	0.62 / 0.60	2.21 (3.50)	0.44 / 0.42	1.06 (1.31)	0.51 / 0.49	1.50 (1.80)
Carry*	0.57 / 0.54	1.93 (3.16)	0.52 / 0.49	1.73 (2.28)	0.49 / 0.47	1.39 (1.78)
FV: inflation + GDP 1y, 10y	0.60 / 0.56	2.13 (3.12)	0.52 / 0.48	1.65 (2.06)	0.50 / 0.47	1.44 (1.68)
FV: fixed in advance	0.50 / 0.47	1.38 (2.47)	0.45 / 0.42	1.21 (1.68)	0.45 / 0.42	1.08 (1.42)
π^*: 3y inflation + r^*	0.54 / 0.52	1.75 (2.31)	0.45 / 0.43	1.15 (1.44)	0.50 / 0.48	1.41 (1.66)
π^*: trend inflation	0.53 / 0.51	1.58 (2.38)	0.49 / 0.48	1.50 (1.65)	0.52 / 0.50	1.53 (1.74)
OLS: future yield	0.61 / 0.59	2.18 (3.50)	0.40 / 0.38	0.74 (0.93)	0.49 / 0.47	1.39 (1.67)
OLS: future yield, cross-section	0.59 / 0.57	2.04 (3.12)	0.39 / 0.37	0.67 (0.90)	0.49 / 0.47	1.32 (1.58)
OLS: future short rate	0.59 / 0.57	2.12 (3.13)	0.44 / 0.42	1.04 (1.43)	0.48 / 0.46	1.33 (1.67)
LASSO: future yield	0.46 / 0.44	1.05 (1.82)	0.42 / 0.41	0.95 (1.48)	0.44 / 0.42	0.89 (1.25)
LASSO: future yield, cross-section	0.47 / 0.45	1.19 (1.98)	0.40 / 0.39	0.75 (1.06)	0.43 / 0.42	0.88 (1.21)
LASSO: future short rate	0.46 / 0.43	1.11 (2.01)	0.36 / 0.34	0.49 (0.71)	0.40 / 0.37	0.65 (0.89)
Elastic net 0.5: future yield	0.49 / 0.47	1.22 (2.42)	0.49 / 0.47	1.38 (2.38)	0.46 / 0.44	1.07 (1.65)
Elastic net 0.5: future yield, cross-section	0.42 / 0.40	0.92 (1.50)	0.45 / 0.43	1.06 (1.71)	0.43 / 0.41	0.86 (1.20)
Elastic net 0.5: future short rate	0.51 / 0.49	1.39 (2.30)	0.40 / 0.38	0.74 (1.04)	0.41 / 0.39	0.77 (1.01)
Elastic net CV: future yield	0.49 / 0.47	1.21 (2.39)	0.49 / 0.46	1.30 (2.26)	0.48 / 0.46	1.18 (1.80)
Elastic net CV: future yield, cross-section	0.51 / 0.49	1.48 (2.43)	0.50 / 0.48	1.48 (2.11)	0.48 / 0.46	1.22 (1.63)
Elastic net CV: future short rate	0.52 / 0.50	1.53 (2.55)	0.50 / 0.48	1.34 (1.85)	0.46 / 0.44	1.05 (1.41)
LASSO / EN: BR structure	0.55 / 0.54	1.72 (2.56)	0.50 / 0.49	1.54 (1.70)	0.53 / 0.51	1.61 (1.80)
Spike-and-slab: future yield	0.41 / 0.39	0.84 (1.36)	0.35 / 0.32	0.40 (0.64)	0.38 / 0.36	0.55 (0.83)
Spike-and-slab: future yield, cross-section	0.48 / 0.46	1.22 (2.11)	0.44 / 0.41	1.09 (1.46)	0.45 / 0.43	1.01 (1.41)
Spike-and-slab: future short rate	0.49 / 0.47	1.32 (2.21)	0.52 / 0.49	1.58 (2.22)	0.49 / 0.47	1.34 (1.92)
Spike-and-slab: BR structure	0.51 / 0.49	1.56 (2.16)	0.53 / 0.52	1.75 (2.44)	0.53 / 0.52	1.70 (2.16)

Panel B: Carry and Static Carry

	Carry SR / net	Static Carry SR
Median (paper)	0.35 / 0.29	0.30
Terciles	0.41 / 0.35	0.31
Rank weights	0.41 / 0.34	0.30

Table D2: Dropping one currency at a time. Range of the Sharpe ratio over the nine samples, the currency whose removal gives the lowest, the number of samples in which the version beats Carry and Static Carry, and in which its alpha over Dollar, Carry and Static Carry is significant at 5%, with the largest p-value. Versions as in Table A1. October 1987 to February 2025.

Version	SR min–max	Lowest without	Beats Carry / Static Carry	α on both significant	Largest p
Hamilton	0.37–0.54	JP	9 / 9 of 9	3 of 9	.472
10-year yield	0.30–0.41	NZ	6 / 8 of 9	0 of 9	.778
FV: inflation 1y+3y	0.48–0.66	GB	9 / 9 of 9	9 of 9	.038
Carry*	0.45–0.65	NZ	9 / 9 of 9	9 of 9	.028
FV: inflation + GDP 1y, 10y	0.48–0.67	GB	9 / 9 of 9	9 of 9	.046
FV: fixed in advance	0.43–0.54	NZ	9 / 9 of 9	5 of 9	.116
π^*: 3y inflation + r^*	0.42–0.59	GB	9 / 9 of 9	4 of 9	.198
π^*: trend inflation	0.39–0.60	SE	9 / 9 of 9	5 of 9	.234
OLS: future yield	0.49–0.63	GB	9 / 9 of 9	9 of 9	.030
OLS: future yield, cross-section	0.50–0.59	CA	9 / 9 of 9	9 of 9	.021
OLS: future short rate	0.47–0.62	GB	9 / 9 of 9	8 of 9	.064
LASSO: future yield	0.40–0.55	CA	8 / 9 of 9	2 of 9	.390
LASSO: future yield, cross-section	0.40–0.48	CA	9 / 9 of 9	2 of 9	.244
LASSO: future short rate	0.29–0.52	EU	7 / 9 of 9	4 of 9	.617
Elastic net 0.5: future yield	0.36–0.60	NZ	9 / 9 of 9	3 of 9	.247
Elastic net 0.5: future yield, cross-section	0.32–0.48	CA	7 / 9 of 9	2 of 9	.492
Elastic net 0.5: future short rate	0.36–0.57	EU	9 / 9 of 9	4 of 9	.273
Elastic net CV: future yield	0.36–0.61	NZ	9 / 9 of 9	4 of 9	.235
Elastic net CV: future yield, cross-section	0.46–0.53	EU	9 / 9 of 9	7 of 9	.081
Elastic net CV: future short rate	0.45–0.57	CA	9 / 9 of 9	7 of 9	.141
LASSO / EN: BR structure	0.42–0.61	SE	9 / 9 of 9	7 of 9	.138
Spike-and-slab: future yield	0.30–0.45	NZ	7 / 9 of 9	0 of 9	.526
Spike-and-slab: future yield, cross-section	0.39–0.59	CA	8 / 9 of 9	3 of 9	.301
Spike-and-slab: future short rate	0.42–0.59	EU	9 / 9 of 9	3 of 9	.208
Spike-and-slab: BR structure	0.42–0.55	EU	9 / 9 of 9	3 of 9	.132

Table D3: Sub-periods for the main versions. Sharpe ratio of each version and of Carry, and the alpha of the version over Dollar, Carry and Static Carry in % a year (Newey-West t-statistic). Versions as in Table A1.

	1987–2005	2006–2025	1987–1999	2000–2012	2013–2025
Carry, Sharpe ratio	0.53	0.18	0.37	0.56	0.07
Carry*, Sharpe ratio	0.82	0.32	0.75	0.68	0.24
Carry*, α (%)	2.95	1.08	3.33	1.40	1.53
	(2.44)	(1.79)	(2.23)	(2.16)	(1.86)
FV: inflation 1y+3y, Sharpe ratio	0.82	0.42	0.67	0.62	0.54
FV: inflation 1y+3y, α (%)	2.94	1.65	2.58	1.21	2.82
	(2.42)	(2.02)	(2.01)	(1.28)	(2.68)
π^*: 3y inflation + r^*, Sharpe ratio	0.70	0.39	0.52	0.54	0.59
π^*: 3y inflation + r^*, α (%)	2.21	1.51	1.61	0.56	2.99
	(1.47)	(1.71)	(0.94)	(0.56)	(2.91)
π^*: trend inflation, Sharpe ratio	0.76	0.30	0.61	0.54	0.40
π^*: trend inflation, α (%)	2.50	0.83	2.24	0.67	2.08
	(1.89)	(1.24)	(1.32)	(0.67)	(2.73)
OLS: future yield, Sharpe ratio	0.82	0.41	0.69	0.61	0.53
OLS: future yield, α (%)	2.95	1.57	2.67	1.14	2.69
	(2.42)	(2.01)	(2.08)	(1.21)	(2.75)
Hamilton, Sharpe ratio	0.84	0.13	0.72	0.59	0.06
Hamilton, α (%)	2.17	−0.31	2.17	0.68	0.20
	(3.07)	(−0.50)	(2.71)	(0.83)	(0.34)
10-year yield, Sharpe ratio	0.65	0.14	0.47	0.55	0.12
10-year yield, α (%)	0.94	−0.31	0.44	0.53	0.57
	(1.25)	(−0.60)	(0.45)	(0.70)	(0.98)

Table D4: Sorting on forward-implied long-horizon differentials. The 1-, 5- and 10-year differentials implied by forward exchange rates, from November 1996, with columns as in Table 10, Panel A. The race uses the portfolios of Della Corte and Riddiough (2025) only, since our own test assets need 120 months with all nine currencies.

	SR	SR net	α on Carry (%)	α on Carry and Static Carry (%)	Race, DCR: factor (t)	Race, DCR: Carry t
1y forward-implied differential	0.32	0.31	0.04	0.14	2.17	1.27
			(0.11)	(0.35)	(1.66)	
5y forward-implied differential	0.33	0.33	0.24	0.41	1.99	1.50
			(0.49)	(0.88)	(1.51)	
10y forward-implied differential	0.34	0.33	0.23	0.37	2.04	1.54
			(0.62)	(0.95)	(1.52)	
10-year yield	0.30	0.29	−0.01	0.17	1.83	1.58
			(−0.02)	(0.39)	(1.39)	
Hamilton	0.32	0.30	0.14	0.29	2.42	1.20
			(0.32)	(0.57)	(1.85)	
Carry*	0.44	0.41	1.07	1.20	2.57	1.32
			(1.95)	(2.47)	(2.00)	

Table D5: Stale signals. Strategies on a signal 0 to 24 months old, from October 1990: Sharpe ratio, and below it the alpha over Dollar, Carry and Static Carry in % a year (Newey-West t-statistic). Versions as in Table A1.

	Lag 0	Lag 1	Lag 3	Lag 6	Lag 12	Lag 24
Carry*	0.52	0.41	0.48	0.25	0.28	0.31
	1.89 (2.87)	1.20 (1.72)	1.67 (2.35)	0.23 (0.29)	0.44 (0.65)	0.67 (0.90)
FV: inflation 1y+3y	0.57	0.52	0.45	0.32	0.26	0.13
	2.17 (3.19)	1.86 (2.88)	1.49 (2.00)	0.74 (0.90)	0.41 (0.59)	−0.25 (−0.32)
π^*: 3y inflation + r^*	0.48	0.33	0.32	0.29	0.21	0.16
	1.64 (2.01)	0.84 (1.04)	0.71 (0.93)	0.56 (0.83)	0.13 (0.17)	−0.04 (−0.05)
π^*: trend inflation	0.48	0.49	0.41	0.31	0.25	0.19
	1.54 (2.16)	1.59 (2.23)	1.25 (1.41)	0.75 (0.84)	0.39 (0.60)	0.02 (0.04)
OLS: future yield	0.56	0.53	0.46	0.34	0.24	0.15
	2.13 (3.18)	1.95 (3.05)	1.56 (2.08)	0.86 (1.07)	0.32 (0.47)	−0.11 (−0.13)
Hamilton	0.40	0.23	0.20	0.11	0.13	0.28
	0.74 (1.45)	−0.35 (−0.55)	−0.59 (−1.05)	−1.18 (−2.33)	−0.96 (−1.69)	0.13 (0.27)
10-year yield	0.33	0.33	0.25	0.22	0.25	0.23
	0.25 (0.54)	0.25 (0.45)	−0.29 (−0.57)	−0.46 (−0.95)	−0.27 (−0.55)	−0.32 (−0.61)
Carry	0.32	0.27	0.19	0.19	0.17	0.05
	−0.00 (−1.65)	0.06 (0.08)	−0.47 (−0.72)	−0.39 (−0.53)	−0.39 (−0.56)	−0.72 (−0.84)

E. Other currency factors

Table E1 extends the value and momentum regressions of Table 7 to every version of the long-run rate, using both our own factors and those of Della Corte and Riddiough (2025). All versions earn a positive abnormal return over value and momentum. For *carry** it is 3.07% per year with our own factors, with a t-statistic of 3.86, and 1.65% when the factors of Della Corte and Riddiough (2025) enter together with carry and static carry, with a t-statistic of 2.99. In this last specification the abnormal return remains significant at 5% for 18 of the 25 versions, including the fair values and the π^* versions, but not for the Hamilton trend or the 10-year yield.

Table E2 looks more closely at the term spread factor of Ang and Chen (2010). The abnormal return of *carry** is 1.93% per year after controlling for the dollar factor, carry, static carry and the term spread factor, with a t-statistic of 3.13, while the term spread factor earns no abnormal return over *carry**. Sorting within each half of the term spread, *carry** still earns a Sharpe ratio of 0.33 with a t-statistic of 2.02. The information in the long-run rate is therefore distinct from the slope of the yield curve. Table E3 does the same for the global imbalance, size and commodity factors. *Carry** keeps an abnormal return of 2.06% per year over the dollar factor, carry, static carry, global imbalance and size, and of 2.57% over carry, static carry and the commodity factor between 1988 and 2013, while none of these factors earns a significant abnormal return over *carry**. Finally, Table E4 shows that countries with high long-run rates tend to be net debtors, small and commodity exporters, with rank correlations of −0.35, −0.24 and 0.53. These links are

weaker than those of the forward premium and much weaker than those of the Hamilton trend and the 10-year yield, whose correlations are −0.62, around −0.48 and around 0.75.

Table E1: Value and momentum for every version. Alpha of the version over Dollar and value and 1-, 3- and 12-month momentum built on our nine currencies (own), as in Table 7, Panel A, alone and with Carry and Static Carry. The other columns use the value factor, or the 1-, 3- and 12-month momentum factors together, of Della Corte and Riddiough (2025) (DCR), quintile sorts on G10 currencies, each with Dollar, and all of them with Carry and Static Carry. Alphas in % a year. Newey-West t-statistics (12 lags) in parentheses. Versions as in Table A1. October 1987 to February 2025.

Version	α on own value and momentum	α on own value and momentum, Carry and Static Carry	α on DCR value	α on DCR momentum	α on DCR value and momentum, Carry and Static Carry
Hamilton	2.29 (2.67)	0.96 (1.94)	2.15 (2.42)	2.80 (3.01)	0.74 (1.52)
10-year yield	1.82 (1.91)	0.44 (1.03)	1.72 (1.73)	2.36 (2.30)	0.24 (0.56)
FV: inflation 1y+3y	3.29 (4.10)	2.37 (3.79)	3.15 (3.70)	3.36 (3.86)	2.04 (3.38)
Carry*	3.07 (3.86)	1.96 (3.44)	3.00 (3.49)	3.27 (3.62)	1.65 (2.99)
FV: inflation + GDP 1y, 10y	3.13 (3.86)	2.10 (3.13)	3.16 (3.66)	3.37 (3.69)	1.93 (2.99)
FV: fixed in advance	2.65 (3.28)	1.61 (2.90)	2.49 (3.08)	2.75 (3.24)	1.30 (2.52)
π^*: 3y inflation + r^*	2.84 (3.27)	1.92 (2.65)	2.70 (2.97)	2.93 (3.04)	1.62 (2.22)
π^*: trend inflation	2.83 (3.39)	1.81 (2.87)	2.65 (2.97)	2.96 (3.14)	1.48 (2.30)
OLS: future yield	3.26 (4.10)	2.35 (3.81)	3.11 (3.69)	3.33 (3.86)	2.01 (3.37)
OLS: future yield, cross-section	3.19 (3.84)	2.26 (3.49)	2.99 (3.45)	3.21 (3.58)	1.89 (2.99)
OLS: future short rate	3.32 (3.97)	2.32 (3.48)	3.13 (3.50)	3.37 (3.65)	1.97 (3.00)
LASSO: future yield	2.42 (2.60)	1.31 (2.22)	2.22 (2.34)	2.59 (2.80)	1.02 (1.78)
LASSO: future yield, cross-section	2.52 (3.10)	1.55 (2.65)	2.30 (2.74)	2.48 (2.94)	1.16 (1.92)
LASSO: future short rate	2.44 (3.07)	1.41 (2.53)	2.25 (2.74)	2.38 (2.84)	1.00 (1.89)
Elastic net 0.5: future yield	2.58 (2.97)	1.47 (2.74)	2.47 (2.80)	2.75 (3.16)	1.23 (2.45)
Elastic net 0.5: future yield, cross-section	2.28 (2.75)	1.28 (2.11)	2.05 (2.39)	2.21 (2.53)	0.87 (1.40)
Elastic net 0.5: future short rate	2.76 (3.31)	1.70 (2.78)	2.51 (2.91)	2.70 (3.12)	1.30 (2.17)
Elastic net CV: future yield	2.63 (3.05)	1.50 (2.77)	2.49 (2.89)	2.78 (3.23)	1.25 (2.46)
Elastic net CV: future yield, cross-section	2.81 (3.50)	1.73 (2.81)	2.69 (3.27)	2.80 (3.34)	1.44 (2.40)
Elastic net CV: future short rate	2.95 (3.36)	1.78 (2.79)	2.82 (3.12)	2.97 (3.29)	1.49 (2.55)
LASSO / EN: BR structure	3.04 (3.51)	2.00 (3.02)	2.80 (3.12)	3.13 (3.30)	1.61 (2.45)
Spike-and-slab: future yield	2.18 (2.52)	1.09 (1.77)	2.05 (2.34)	2.25 (2.61)	0.78 (1.34)
Spike-and-slab: future yield, cross-section	2.68 (3.10)	1.52 (2.63)	2.56 (2.91)	2.70 (3.01)	1.25 (2.26)
Spike-and-slab: future short rate	2.73 (3.03)	1.59 (2.58)	2.59 (2.84)	2.78 (2.96)	1.29 (2.25)
Spike-and-slab: BR structure	2.99 (3.41)	1.89 (2.64)	2.72 (3.00)	2.97 (3.10)	1.43 (2.07)

Table E2: The term spread factor. Panel A: alpha of the version over Dollar, Carry, Static Carry and the Ang and Chen (2010) term spread factor, the alpha of the term spread factor over the version (and Carry and Static Carry), in % a year, and the version's Sharpe ratio within term spread halves. Panel B: the factors, with $i^* - \text{fp}$ the gap between the $Carry^*$ differential and the forward premium. Newey-West t-statistics (12 lags) in parentheses. Versions as in Table A1. October 1987 to February 2025.

Panel A: Alphas Version	α on Dollar + Carry + Static Carry + slope	Slope's α on version + Dollar	Slope's α on version + Carry + Static Carry	Version within slope halves, SR (t)
Carry*	1.93	0.25	0.23	0.33
	(3.13)	(0.26)	(0.25)	(2.02)
FV: inflation 1y+3y	2.22	0.34	0.35	0.43
	(3.52)	(0.35)	(0.38)	(2.81)
π^*: 3y inflation + r^*	1.76	0.30	0.38	0.42
	(2.36)	(0.30)	(0.41)	(2.55)
π^*: trend inflation	1.55	−0.01	0.14	0.44
	(2.31)	(−0.01)	(0.14)	(2.71)
OLS: future yield	2.19	0.32	0.35	0.42
	(3.52)	(0.33)	(0.38)	(2.71)
Hamilton	0.88	−0.08	0.06	0.25
	(1.81)	(−0.08)	(0.07)	(1.64)
10-year yield	0.35	−0.19	0.17	0.32
	(0.77)	(−0.19)	(0.18)	(2.00)
Panel B: Factors Factor	Mean (%)	SR	t	
Term spread	−0.30	−0.05	−0.30	
$i^* - \text{fp}$	−0.97	−0.16	−0.91	
Carry*	3.52	0.57	3.57	
Carry	2.18	0.35	2.43	
Static Carry	1.84	0.30	1.79	

Table E3: Global imbalance, size and commodity factors. Alpha of the version over the global imbalance (Della Corte et al., 2016), size (Hassan, 2013) and commodity (Ready et al., 2017) factors, and of each factor over the version and Dollar, in % a year. Net foreign assets from Lane and Milesi-Ferretti (2018). Panel B: the factors. Newey-West t-statistics (12 lags) in parentheses. Versions as in Table A1. October 1987 to February 2025.

Panel A: Alphas Version	α on Dollar + Carry + Static Carry + imbalance + size	α on Carry + Static Carry + commodity (1988–2013)	Imbalance's α on version	Size's α on version	Commodity's α on version
Carry*	2.06	2.57	0.20	0.59	1.83
	(3.49)	(3.34)	(0.26)	(1.03)	(1.19)
FV: inflation 1y+3y	2.32	2.04	0.27	0.66	1.84
	(3.72)	(2.46)	(0.34)	(1.12)	(1.19)
π^*: 3y inflation + r^*	1.84	1.25	0.36	0.71	2.18
	(2.44)	(1.21)	(0.49)	(1.22)	(1.53)
π^*: trend inflation	1.71	1.51	0.11	0.70	1.67
	(2.54)	(1.56)	(0.15)	(1.25)	(1.19)
OLS: future yield	2.30	2.04	0.29	0.66	1.83
	(3.75)	(2.43)	(0.37)	(1.13)	(1.19)
Hamilton	1.07	1.41	−0.29	0.60	0.72
	(2.36)	(2.42)	(−0.39)	(1.01)	(0.53)
10-year yield	0.60	0.44	0.01	0.79	1.32
	(1.52)	(0.76)	(0.02)	(1.37)	(1.02)

Panel B: Factors Factor	Mean (%)	SR	t
Global imbalance (net debtors long)	1.87	0.29	1.86
Size (small long)	1.31	0.26	1.62
Commodity (RRW, 1988–2013)	4.61	0.53	2.82

Table E4: Cross-country links. Mean monthly cross-sectional rank correlation of each version's differential with net foreign assets over GDP (Lane and Milesi-Ferretti, 2018), country size, and the finished-goods imports minus exports ratio of Ready et al. (2017) (1988 to 2013), high for commodity exporters. Versions as in Table A1.

Version	NFA / GDP	Size	Finished-goods imports - exports (RRW imx)
Carry*	−0.35	−0.24	0.53
FV: inflation 1y+3y	−0.34	−0.17	0.55
π^*: 3y inflation + r^*	−0.35	−0.17	0.55
π^*: trend inflation	−0.35	−0.21	0.54
OLS: future yield	−0.35	−0.17	0.55
Hamilton	−0.62	−0.47	0.75
10-year yield	−0.62	−0.48	0.76
Forward premium	−0.41	−0.35	0.57

F. Additional results on the source of the premium

We first report the uncovered interest parity tests behind Table 9 in more detail. Table F1 reports Fama regressions of the monthly depreciation on the forward premium. The slope is 0.68 in the pooled regression, 0.75 with currency effects and 0.69 with month effects, and none of these slopes is significantly different from one. When the forward premium is split into its currency mean, its month mean and the rest, as in Hassan and Mano (2019), the slope is lowest on the currency mean, 0.30 with a t-statistic against one of −1.83, which is the part closest to the signal of static carry. Table F2 repeats the split of the forward premium into its long-run part and the rest of Table 9 for the main versions of the long-run rate. The result is the same for all of them. The slope on the long-run part is negative in the pooled regression, between −0.82 and −0.06, and below one with t-statistics between −3.82 and −2.73, and it stays far below one with currency effects and with month effects. The slope on the rest is between 0.88 and 1.16 and never significantly different from one. Figure F1 splits the return of every strategy into the interest differential and the currency movement. The interest differential earned is similar across the versions of the long-run rate, between 2.25% and 2.90% per year, and most of the difference in their returns comes from the currency movement. The currency movement adds between 0.53% and 1.28% per year for the fair values and between 0.96% and 1.12% for the OLS versions, while it subtracts 1.96% for carry and 0.91% for static carry.

The next tables concern the inflation part of Subsection 4.1. Table F3 adds a country r^* to π^*, as in the decomposition of Bauer and Rudebusch (2020). With r^* from 5- or 10-year GDP growth or from the 5-year real short rate, the Sharpe ratio falls from 0.53 and 0.54 to between 0.41 and 0.46, and the abnormal return over carry and static carry is no longer significant, except with 3-year inflation and 10-year GDP growth. Table F4 compares our real-time π^* with market measures of long-run inflation. The cross-sectional correlation of the differentials to the US is 0.93 with 5-year, 5-year inflation swaps and between 0.55 and 0.67 with breakevens, and the level of π^* is close to that of the swaps, with a mean gap of 0.15 and 0.04 percentage points. The π^* differential also predicts the inflation differential over the next five years, with slopes of 0.30 and 0.32 and t-statistics of 2.87 and 2.74. Table F5 relates the returns to inflation news. The sorts on π^* gain when the US 10-year breakeven rises, with slopes of 44.69 and 30.37 and t-statistics of 4.56 and 3.33, and keep a significant exposure once carry is controlled for. The sorts on the real part have smaller slopes, which become close to zero once carry is controlled for. The 10-year yield and static carry have a similar exposure, and no strategy has a significant exposure to the 5-year, 5-year inflation swap once carry is controlled for. Finally, Table F6 splits the

forward premium into the parts explained by the π^* differential and by the real part. The slope on the inflation part is between -0.65 and 0.01, below one with t-statistics between -3.75 and -2.88, while the slopes on the real part and on the rest are not significantly different from one. Exchange rates therefore offset the real part of the interest differential but not the inflation part, which is why the premium of *carry** is in the inflation part.

Table F1: Fama regressions. Each column regresses each currency's depreciation against the US dollar over the next month, $\Delta s_{c,t+1}$, on its forward premium, both in % a year, pooling the nine currencies. Column 1 is pooled. Column 2 demeans every variable by currency, so the slope uses only the time series of each currency, and column 3 by month, so it uses only the differences across currencies. Column 4 splits the forward premium into its currency mean, its month mean and the rest (Hassan and Mano, 2019). Columns 1 and 3 are columns 1 and 2 of Table 9. Driscoll-Kraay t-statistics of the slope against one in parentheses. October 1987 to February 2025.

	Dependent variable: $\Delta s_{c,t+1}$			
	(1)	(2)	(3)	(4)
Forward premium	0.68	0.75	0.69	
	(-1.18)	(-0.85)	(-1.52)	
Currency mean				0.30
				(-1.83)
Month mean				0.66
				(-0.63)
Rest				0.81
				(-0.92)
Currency effects	No	Yes	No	No
Month effects	No	No	Yes	No

Table F2: Uncovered interest parity for the long-run part, every main version. Each row regresses each currency's depreciation against the US dollar over the next month, $\Delta s_{c,t+1}$, on the long-run part of its forward premium, the projection on that version's i^* differential, and the rest, as in Table 9. The currency effects column demeans every variable by currency and reports only the slope on the long-run part. The month effects columns demean every variable by month. Driscoll-Kraay t-statistics of the slope against one in parentheses. The last column is the correlation of the forward premium with i^*. Versions as in Table A1. October 1987 to February 2025.

	Dependent variable: $\Delta s_{c,t+1}$					
	Pooled		Currency effects	Month effects		
	Long-run part	Rest	Long-run part	Long-run part	Rest	Corr(fp, i^*)
Hamilton	−0.06	1.09	−0.42	0.12	0.98	0.60
	(−2.73)	(0.31)	(−2.39)	(−2.62)	(−0.10)	
10-year yield	−0.13	1.16	−0.55	0.23	0.94	0.61
	(−3.20)	(0.59)	(−2.83)	(−2.27)	(−0.30)	
FV: inflation 1y+3y	−0.78	1.05	−1.94	−0.13	0.92	0.45
	(−3.82)	(0.18)	(−3.15)	(−3.29)	(−0.43)	
Carry*	−0.64	1.00	−1.65	−0.07	0.88	0.45
	(−3.36)	(0.01)	(−2.57)	(−2.84)	(−0.61)	
FV: inflation + GDP 1y, 10y	−0.78	1.04	−1.73	−0.13	0.88	0.45
	(−3.43)	(0.14)	(−2.70)	(−2.89)	(−0.61)	
π^*: 3y inflation + r^*	−0.44	0.99	−1.16	−0.02	0.92	0.47
	(−3.21)	(−0.03)	(−2.49)	(−2.89)	(−0.44)	
π^*: trend inflation	−0.78	1.10	−1.81	−0.05	0.93	0.47
	(−3.59)	(0.32)	(−3.03)	(−3.10)	(−0.38)	
OLS: future yield	−0.82	1.04	−2.09	−0.11	0.90	0.44
	(−3.75)	(0.13)	(−3.10)	(−3.23)	(−0.50)	

Table F3: Bauer-Rudebusch with a country r^*. $i^* = \pi^* + r^*$ with r^* from 5- or 10-year GDP growth or the 5-year real short rate, next to a common r^*, with columns as in Table 10, Panel A. π^* is one of the two real-time measures of Table 10: trend inflation, the discounted average of past 12-month CPI inflation in equation (5), with the discount chosen each month in real time, or CPI inflation over the past three years. October 1987 to February 2025.

	SR	SR net	α on Carry (%)	α on Carry and Static Carry (%)	Race, own: factor (t)	Race, own: Carry t	Race, DCR: factor (t)	Race, DCR: Carry t
Trend inflation, common r^*	0.53	0.51	1.84	1.58	2.40	1.71	3.13	1.95
			(2.35)	(2.38)	(2.32)		(2.88)	
Trend inflation + 5-year GDP growth	0.42	0.40	1.16	0.81	1.86	1.95	2.93	1.94
			(1.54)	(1.28)	(1.69)		(2.54)	
Trend inflation + 10-year GDP growth	0.42	0.41	1.17	0.78	1.85	1.88	2.83	1.94
			(1.68)	(1.46)	(1.73)		(2.56)	
Trend inflation + 5-year real short rate	0.41	0.40	0.95	0.53	1.63	1.29	2.68	1.92
			(1.26)	(0.97)	(1.58)		(2.51)	
3-year inflation, common r^*	0.54	0.52	1.93	1.75	2.36	1.63	3.20	1.75
			(2.27)	(2.31)	(2.40)		(2.92)	
3-year inflation + 10-year GDP growth	0.46	0.45	1.47	1.20	1.96	1.91	3.13	1.79
			(2.03)	(2.05)	(1.82)		(2.73)	
3-year inflation + 5-year real short rate	0.44	0.43	1.13	0.75	1.50	1.53	2.69	1.92
			(1.53)	(1.30)	(1.47)		(2.56)	
10-year yield	0.39	0.37	0.84	0.38	1.61	2.21	2.57	2.16
			(1.35)	(0.87)	(1.42)		(2.18)	
Hamilton	0.47	0.44	1.34	0.91	2.01	1.80	2.92	1.98
			(2.11)	(1.83)	(1.81)		(2.56)	
Carry*	0.57	0.54	2.16	1.93	2.87	1.36	3.78	1.48
			(2.94)	(3.16)	(2.67)		(3.30)	

Table F4: Validating the real-time π^*. Panel A compares the two real-time measures of π^* of Table 10, trend inflation and 3-year CPI inflation, with market measures of long-run inflation from LSEG: 5-year, 5-year forward inflation swaps (US, GB, JP, AU, EU) and 10-year and 5-year, 5-year forward breakevens, to February 2025. The rows give the pooled correlation of levels, the mean gap (market minus real-time, percentage points), the mean monthly cross-sectional correlation of the differentials to the US, and the slope of the market differential on the real-time one, with its Driscoll-Kraay standard error in parentheses. In Panel B, columns 1 and 2 regress the realised inflation differential to the US over the next five years on the π^* differential and the real part, and columns 3 and 4 regress the depreciation against the US dollar over the next five years on the realised inflation differential (relative purchasing power parity), alone and with the forward premium. Driscoll-Kraay t-statistics against zero in parentheses.

Panel A: Market long-run inflation

	5-year, 5-year inflation swap		10-year breakeven		5-year, 5-year breakeven	
	Trend inflation	3-year inflation	Trend inflation	3-year inflation	Trend inflation	3-year inflation
Countries	5	5	8	8	8	8
From	Jun 2007	Jun 2007	Sep 2004	Sep 2004	Oct 2004	Oct 2004
Correlation of levels	0.59	0.56	0.48	0.48	0.34	0.33
Mean gap (pp)	0.15	0.04	−0.32	−0.42	−0.26	−0.36
Cross-sectional correlation	0.93	0.93	0.67	0.61	0.59	0.55
Slope (s.e.)	0.76	0.76	0.42	0.38	0.43	0.41
	(0.09)	(0.08)	(0.10)	(0.08)	(0.08)	(0.07)

Panel B: Realised inflation

	Future inflation differential		Future depreciation	
	(1)	(2)	(3)	(4)
π^* differential	0.30	0.32		
	(2.87)	(2.74)		
Real part	0.23	0.22		
	(3.51)	(3.78)		
Realised inflation differential			0.69	0.61
			(1.86)	(1.56)
Forward premium				0.08
				(1.47)
π^* measure	Trend	3-year		
R^2	0.25	0.25	0.03	0.03
Months	409	409	390	390

Table F5: Exposure to inflation news. Each cell is the slope of a regression of the monthly return of a strategy (% a year) on a measure of inflation news over the same month, in percentage points, so it gives the return per year for a one percentage point rise in the measure. The measures are the change in the US 10-year breakeven inflation rate (from September 2004, 246 months), the change in the US 5-year, 5-year forward inflation swap rate (from June 2007, 213 months), both from LSEG, and the change in the mean real-time 12-month CPI inflation of the ten countries (October 1987 to February 2025, 449 months). Columns marked alone control for Dollar, and columns marked with Carry control for Dollar and Carry, so they ask whether the strategy is exposed to inflation news beyond what carry is. The rows are the sorts on the π^* differential and on the real part, the yield minus π^*, with π^* measured by trend inflation or 3-year inflation as in Table 10, then $Carry^*$, the 10-year yield, Carry and Static Carry. Newey-West t-statistics (12 lags) in parentheses.

	Δ US 10y breakeven: alone	Δ US 10y breakeven: with Carry	Δ US 5y5y inflation swap: alone	Δ US 5y5y inflation swap: with Carry	Δ mean 12m inflation, all countries: alone	Δ mean 12m inflation, all countries: with Carry
π^*: trend inflation	44.69	24.17	29.84	6.59	−4.84	−1.40
	(4.56)	(3.92)	(1.45)	(0.41)	(−1.26)	(−0.46)
Real part (trend)	17.19	3.71	6.81	−10.36	−5.69	−2.89
	(1.97)	(0.41)	(0.47)	(−0.83)	(−1.57)	(−0.94)
π^*: 3-year inflation	30.37	12.84	16.46	−1.71	−0.64	2.49
	(3.33)	(2.15)	(1.15)	(−0.14)	(−0.17)	(0.81)
Real part (3-year)	15.30	2.22	12.45	−3.93	−5.91	−3.15
	(1.85)	(0.29)	(1.38)	(−0.59)	(−1.38)	(−0.84)
$Carry^*$	37.46	17.85	22.11	−0.22	−5.08	−1.37
	(3.74)	(3.03)	(1.14)	(−0.01)	(−1.28)	(−0.42)
10-year yield	52.17	26.80	36.94	6.90	−7.62	−3.00
	(4.71)	(4.53)	(1.46)	(0.38)	(−1.81)	(−0.99)
Carry	34.41		38.81		−6.17	
	(2.66)		(3.05)		(−1.40)	
Static Carry	38.49	21.67	20.90	−1.81	−8.99	−5.64
	(4.76)	(3.96)	(1.34)	(−0.17)	(−2.31)	(−1.92)

Table F6: Uncovered interest parity for the inflation part and the real part. The table asks which part of the interest differential is offset by currency movement. We first split the 10-year yield differential to the US of each currency into the π^* differential and the real part, the yield differential minus the π^* differential, as in Table 10. We then project the forward premium on these two differentials in a pooled regression over the full sample, and split it into three pieces that add up to the forward premium: the inflation part, the fitted contribution of the π^* differential, the real part, the fitted contribution of the real part, and the rest, the intercept plus the residual. Each column regresses each currency's depreciation against the US dollar over the next month, $\Delta s_{c,t+1}$, in % a year, on the three pieces, pooling the nine currencies. Under uncovered interest parity each slope is one, so the piece is fully offset by depreciation, while a slope below one means that the piece earns an excess return. Columns 1 and 2 measure π^* by trend inflation and columns 3 and 4 by 3-year inflation. In columns 2 and 4 every variable, including those of the projection, is demeaned each month, so the slopes use only the differences across currencies. Driscoll-Kraay t-statistics of the slope against one (6 lags) in parentheses. October 1987 to February 2025.

	Dependent variable: $\Delta s_{c,t+1}$			
	π^*: trend inflation		π^*: 3-year inflation	
	(1)	(2)	(3)	(4)
Inflation part	−0.65	−0.03	−0.38	0.01
	(−3.75)	(−3.07)	(−3.46)	(−2.88)
Real part	0.61	0.68	0.19	0.61
	(−0.59)	(−0.64)	(−1.29)	(−0.80)
Rest	1.22	0.97	1.20	0.97
	(0.78)	(−0.16)	(0.71)	(−0.14)
Month effects	No	Yes	No	Yes

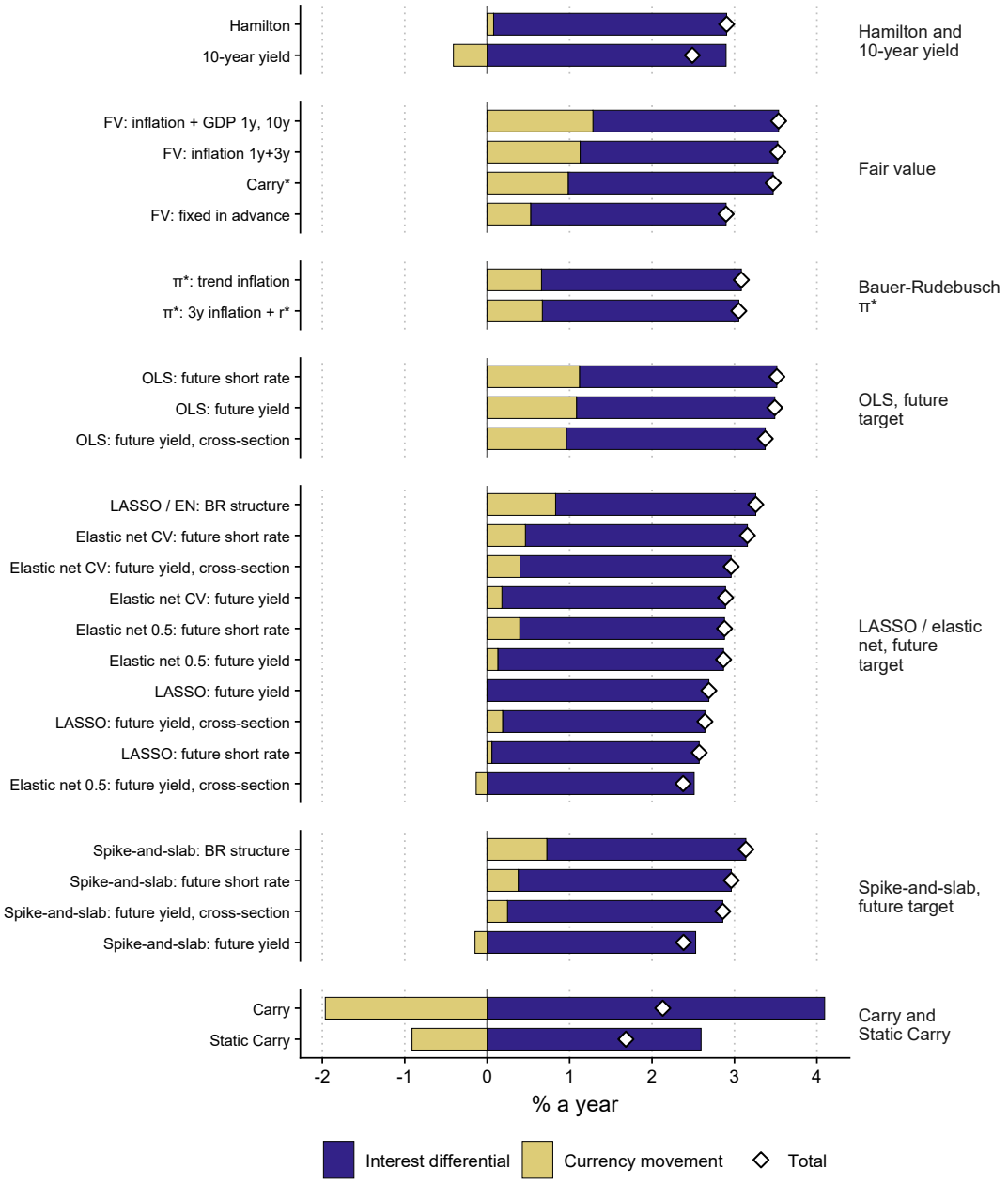


Figure F1: Interest differential and currency movement of the long-short returns for every strategy. Bars split the mean log excess return into the interest differential earned and the currency movement. Diamonds mark the total. Versions are grouped by how i^* is measured (Table A1). Sample October 1987 to February 2025, in % a year.

G. Additional results on risk

Table G1 reports the exposure of the main versions to the US stock market and to global volatility. $Carry^*$ has a CAPM beta of 0.12, below the 0.14 of carry and the 0.18 of static carry, and a downside beta of 0.27 in months when the market falls by more than one standard deviation below its mean, against 0.37 and 0.36. Its CAPM abnormal return is 2.50% per year with a t-statistic of 2.56, and the other macro versions earn between 2.09% and 2.68%, while the abnormal returns of the Hamilton trend, the 10-year yield, carry and static carry are not significant. All strategies lose money when the VIX and global FX volatility rise, with slopes for $carry^*$ that are similar to or smaller than those of the benchmarks. Table G2 reports the cumulative return over nine crash episodes. $Carry^*$ loses money in eight of them, as do carry in seven and static carry in all nine. In the largest episodes its losses are similar to or smaller than those of the benchmarks, −12.03% around the Lehman collapse against −12.71% for carry and −19.63% for static carry, and −5.88% during Covid against −6.28% for both. It loses more than carry during the ERM crisis, when the Swiss National Bank ended the floor and during the 2022 dollar rally.

Table G3 adds the exposure to the option-implied measures at formation and the abnormal return over carry and a factor sorted on each measure. The long minus short difference in risk reversals is 0.73 for $carry^*$, close to the 0.68 of carry, so $carry^*$ is long the currencies for which protection against a fall is dearer, and this exposure is not significantly different from that of carry. Static carry, in contrast, is long currencies with high implied volatility, with a difference of 1.82, while $carry^*$ shows none. The abnormal return of $carry^*$ over carry and the volatility factor is 1.89% per year with a t-statistic of 2.70. Over carry and the risk reversal factor it is 1.45%, with a t-statistic of 1.38, in a sample that starts only in July 2010. Table G4 turns to oil risk from August 2006. $Carry^*$ earns −10.79% per year in months of oil volatility spikes and 5.01% in other months, a contrast similar to that of carry, static carry and the Hamilton trend. Adding the oil measures as controls barely changes its abnormal return over carry, 1.49% against 1.53%, with a t-statistic of 1.73 in both cases. Finally, $carry^*$ tilts towards the commodity currencies, the Australian dollar, the Canadian dollar, the Norwegian krone and the New Zealand dollar, with a net weight of 0.51, less than carry with 0.64 and the Hamilton trend with 0.77. Its exposure to oil therefore does not come from a larger tilt to commodity currencies.

Table G1: Risk exposures. CAPM beta and alpha on the US market (Fama-French), downside beta (market below its mean minus one standard deviation), and slopes on AR(1) innovations in the VIX and in global FX volatility (from October 1989). Newey-West t-statistics (12 lags) in parentheses. Versions as in Table A1. October 1987 to February 2025.

	Carry*	FV: inflation 1y+3y	π^*: 3y inflation + r^*	π^*: trend inflation	OLS: future yield	Hamilton	10-year yield	Carry	Static Carry
CAPM beta	0.12	0.10	0.10	0.12	0.10	0.17	0.18	0.14	0.18
	(4.52)	(4.01)	(3.91)	(4.60)	(3.90)	(8.08)	(8.35)	(5.10)	(6.76)
Downside beta (market < mean − 1 sd)	0.27	0.22	0.23	0.24	0.22	0.28	0.32	0.37	0.36
	(3.93)	(2.42)	(2.23)	(2.26)	(2.48)	(2.84)	(3.83)	(4.24)	(3.95)
CAPM alpha (% a year)	2.50	2.68	2.19	2.09	2.65	1.45	0.97	0.93	0.22
	(2.56)	(2.82)	(2.12)	(2.09)	(2.81)	(1.46)	(0.91)	(0.96)	(0.21)
VIX innovation (% per point)	−0.14	−0.13	−0.13	−0.14	−0.13	−0.18	−0.18	−0.15	−0.18
	(−8.08)	(−6.35)	(−5.86)	(−6.88)	(−6.47)	(−9.29)	(−9.56)	(−4.63)	(−8.19)
Global FX vol innovation (% per bp)	−0.05	−0.04	−0.04	−0.05	−0.04	−0.05	−0.06	−0.05	−0.05
	(−5.11)	(−3.88)	(−3.31)	(−4.81)	(−3.98)	(−5.95)	(−5.97)	(−4.00)	(−4.62)

Table G2: Crash episodes. Cumulative return in % over each episode. Versions as in Table A1.

Episode	Months	Carry*	FV: inflation 1y+3y	π^*: 3y inflation + r^*	π^*: trend inflation	OLS: future yield	Hamilton	10-year yield	Carry	Static Carry
ERM crisis	1	−3.09	−1.96	−1.96	−1.71	−1.96	−3.50	−3.44	1.20	−1.03
LTCM / yen unwind	3	−7.25	−3.95	−4.28	−3.15	−3.95	−5.04	−6.27	−8.42	−6.04
Lehman	4	−12.03	−15.54	−13.56	−16.04	−16.16	−16.00	−18.95	−12.71	−19.63
Euro crisis	2	−2.36	−2.15	−2.15	−2.14	−2.15	−2.40	−2.14	−3.32	−2.86
SNB ends the floor	1	−4.41	−2.19	−2.19	−4.41	−2.19	−5.31	−5.31	−1.14	−3.76
Brexit vote	1	2.45	4.90	4.90	−2.25	4.90	−1.51	−1.51	−1.23	−0.61
Covid	2	−5.88	−5.27	−5.27	−6.02	−5.27	−7.84	−7.84	−6.28	−6.28
2022 dollar rally	8	−1.51	3.45	4.20	0.78	2.29	1.42	2.28	0.30	−2.60
Yen carry unwind	2	−1.23	−0.85	−0.85	−1.89	−0.85	−3.01	−3.01	−2.67	−2.79

Table G3: Crash risk and volatility from FX options. G10 OTC surfaces from LSEG. Exposure is the long-minus-short mean of the measure at formation, also net of Carry's. The risk reversal is signed so that higher means a dearer put on the foreign currency. β is the slope on the change in the cross-country mean over the holding month, and the last column the alpha over Carry and a factor sorted on the measure (% a year). t-statistics in parentheses.

Measure	From	Signal	Exposure (t)	Minus Carry (t)	β to Δ measure (t)	Skewness	α on measure factor and Carry (% a year, t)
25d risk reversal	Jul 2010	Carry*	0.73	0.10	−0.21	−0.21	1.45
			(8.55)	(1.05)	(−3.86)		(1.38)
	Jul 2010	π^*: trend inflation	0.73	0.05	−0.16	−0.41	2.13
			(10.09)	(0.62)	(−2.45)		(2.32)
	Jul 2010	Hamilton	0.83	0.15	−0.19	−0.37	0.46
			(11.22)	(2.19)	(−2.65)		(0.84)
	Aug 2010	Carry	0.68		−0.08	−0.10	
			(11.47)		(−1.59)		
	Dec 2012	Static Carry	0.93	0.26	−0.26	−0.24	0.43
			(11.01)	(3.18)	(−4.46)		(0.63)
1M ATM vol	Jan 1995	Carry*	0.01	−0.21	−0.06	−0.57	1.89
			(0.04)	(−1.09)	(−9.74)		(2.70)
	Jan 1995	π^*: trend inflation	−0.12	−0.34	−0.06	−0.61	2.29
			(−0.39)	(−1.59)	(−8.52)		(3.01)
	Jan 1995	Hamilton	0.23	0.01	−0.07	−0.41	0.91
			(0.74)	(0.05)	(−9.10)		(1.58)
	Jan 1995	Carry	0.22		−0.06	−0.79	
			(0.86)		(−8.94)		
	Jan 1995	Static Carry	1.82	1.60	−0.07	−0.46	0.56
			(6.56)	(7.01)	(−10.32)		(0.93)

Table G4: Oil risk. From August 2006: slopes on changes in WTI implied volatility and skew and on the oil return, mean return in oil-volatility spikes and other months, alpha over Carry with and without oil controls (% a year, t), and the commodity tilt, the weight on AU, CA, NO and NZ in the long minus the short leg. Oil data from LSEG.

Signal	β Δ oil IV (t)	β Δ oil skew (t)	β oil return (t)	Oil-vol spike months / other (% a year)	α on Carry (% a year, t)	α with oil controls (% a year, t)	Commodity tilt
Carry*	−0.50	−0.67	0.64	−10.79 / 5.01	1.53	1.49	0.51
	(−3.22)	(−2.89)	(4.31)		(1.73)	(1.73)	
π^*: trend inflation	−0.50	−1.00	0.77	−9.63 / 4.55	1.24	1.21	0.47
	(−3.09)	(−4.25)	(4.40)		(1.47)	(1.47)	
Hamilton	−0.54	−0.86	0.77	−12.29 / 3.96	0.12	0.09	0.77
	(−3.80)	(−4.41)	(5.38)		(0.17)	(0.14)	
Carry	−0.21	−0.30	0.46	−9.88 / 3.74		1.41	0.64
	(−1.26)	(−1.07)	(2.98)			(1.40)	
Static Carry	−0.41	−0.70	0.56	−15.35 / 4.65	0.70	0.68	0.50
	(−3.78)	(−3.88)	(3.97)		(0.99)	(1.00)	

H. Time-series strategies and the dollar

The strategies in the main text are cross-sectional. They go long the half of the currencies with the highest signals and short the other half, so they always hold as many currencies long as short and have no net exposure to the US dollar. A natural alternative is a time-series strategy, which compares the signal of each currency with that of the US. Each month, the time-series version holds a currency long when its long-run rate is above the US long-run rate and short otherwise, with a weight of $2/N_t$, the same size as in the median split. Unlike the cross-sectional strategy, it can be net long or net short the dollar. To separate the two, we split its monthly return into a dollar part and the rest. The dollar part is the sum of the portfolio weights times the mean return of all currencies, which is the return of the net position against the dollar. The rest comes from the currencies the strategy holds long relative to those it holds short.

Table H1 reports the results. The time-series version of $carry^*$ earns 4.81% per year with a t-statistic of 2.77 and a Sharpe ratio of 0.47, below the 0.57 of the cross-sectional strategy. It is net short foreign currencies, with a mean net position of -0.42, and is net long foreign currencies in only 19% of months, since the US long-run rate is above that of most of the other countries in most months. Its return is therefore partly a bet on the dollar. The dollar part earns 2.49% per year but is not statistically significant, with a t-statistic of 1.51, while the rest earns 2.32% per year with a t-statistic of 3.13 and has a correlation of 0.71 with the cross-sectional strategy. Over the cross-sectional strategy and the dollar factor, the time-series version earns an alpha of 3.11% per year with a t-statistic of 1.95, significant only at the 10% level. The other fair value versions and the OLS fit to the future yield behave in the same way, while for trend inflation the dollar part is significant. The time-series version of carry, in contrast, is close to neutral on the dollar on average, with a mean net position of -0.07, and that of static carry is net long foreign currencies in 97% of months. The significant part of the return of $carry^*$ therefore comes from ranking currencies against each other, and not from its position against the dollar.

A related question is whether the long-run rate times the dollar itself. Similar to the dollar carry trade of Lustig et al. (2014), which conditions on the average forward premium, the dollar timing strategy holds all currencies long against the US dollar when the mean long-run differential across currencies is positive and short otherwise. We also regress the return of the dollar factor over the next month and the next 12 months on the mean long-run differential. Since the mean differential is persistent, we report, together with Newey-West t-statistics, the p-values of the IVX test of Kostakis et al. (2015), which is robust to persistent predictors. We also report the out-of-sample R^2 of expanding-window

forecasts from October 1992 against the historical mean, with the p-values of Clark and West (2007).

Table H2 shows that the mean long-run differential has some power to predict the dollar. For $carry^*$, the slope at one month is 4.79 with a t-statistic of 1.98 and an IVX p-value of 0.041, and at 12 months it is 3.93 with an IVX p-value of 0.051. The slope is 5.22 next to the mean forward premium, with an IVX p-value of 0.072. The mean differential has a first-order autocorrelation of 0.90, but its innovations are almost uncorrelated with the residuals of the dollar return, with a correlation of 0.06, so its persistence does little to bias the slope. The out-of-sample R^2 is small, 0.31%, with a Clark-West p-value of 0.078. The mean forward premium, in contrast, does not predict the dollar in our sample, with an IVX p-value of 0.392 and a negative out-of-sample R^2. This predictability, however, does not make dollar timing on $carry^*$ profitable. The strategy is long foreign currencies in only 20% of months and earns a Sharpe ratio of 0.15, with an alpha over the dollar factor of 2.70% per year and a t-statistic of 1.33. Dollar timing on the 10-year yield, the Hamilton trend and trend inflation does earn a significant alpha, but for $carry^*$ the evidence on the dollar is weak, which confirms that its premium is cross-sectional.

Table H1: Time-series strategies. Each currency is held long when its signal is above zero (above the US) and short otherwise, weight 2/N. Net position is the share long minus the share short. The dollar part (the sum of the weights, twice the net position, times the mean currency return) and the rest add up to the time-series return (% a year, Newey-West t). The alpha is over the cross-sectional strategy and Dollar (Newey-West t). Versions as in Table A1. October 1987 to February 2025.

Signal	Cross-section: Sharpe (net)	Time series: mean (t)	Sharpe (net)	Net position	Share of months net long	Dollar part (t)	Rest (t)	Corr. of rest with cross-section	Alpha over cross-section and Dollar
Carry*	0.57	4.81	0.47	−0.42	0.19	2.49	2.32	0.71	3.11
	(0.54)	(2.77)	(0.46)			(1.51)	(3.13)		(1.95)
FV: inflation 1y+3y	0.62	4.71	0.48	−0.39	0.20	2.33	2.38	0.71	3.12
	(0.60)	(2.83)	(0.47)			(1.52)	(3.29)		(1.73)
π^*: 3y inflation + r^*	0.54	4.15	0.44	−0.38	0.20	2.26	1.89	0.74	2.83
	(0.52)	(2.62)	(0.43)			(1.51)	(2.60)		(1.72)
π^*: trend inflation	0.53	5.20	0.54	−0.33	0.24	3.81	1.39	0.74	3.79
	(0.51)	(3.14)	(0.53)			(2.50)	(1.85)		(1.93)
OLS: future yield	0.61	5.03	0.53	−0.38	0.21	2.45	2.58	0.72	3.47
	(0.59)	(3.08)	(0.51)			(1.63)	(3.56)		(1.92)
Hamilton	0.47	2.88	0.32	−0.08	0.44	1.65	1.23	0.83	0.96
	(0.44)	(1.80)	(0.30)			(1.29)	(1.35)		(0.59)
10-year yield	0.39	4.34	0.55	−0.08	0.42	2.78	1.56	0.85	2.96
	(0.37)	(3.13)	(0.54)			(2.42)	(1.78)		(2.35)
Carry	0.35	4.05	0.41	−0.07	0.46	2.30	1.74	0.71	2.52
	(0.29)	(2.53)	(0.39)			(1.62)	(2.34)		(1.79)
Static Carry	0.30	1.79	0.21	0.32	0.97	0.16	1.63	0.86	−0.20
	(0.30)	(1.29)	(0.21)			(0.18)	(1.61)		(−0.52)

Table H2: Timing the dollar. Dollar timing holds all currencies long when the mean signal is above zero and short otherwise, with its alpha over Dollar in % a year (Newey-West t). Predictive regressions of the Dollar return over 1 or 12 months on the mean signal, with Newey-West t and IVX p-values (Kostakis et al., 2015), out-of-sample R^2 from October 1992 (Clark-West p, Clark and West, 2007), and the slope next to the mean forward premium. AR(1) is the signal's persistence and δ the correlation of its innovations with return residuals. Versions as in Table A1.

Signal	Dollar timing: Sharpe (net)	Share of months long foreign currencies	Alpha over Dollar	Slope, 1 month (t)	IVX p	R^2 %	Out-of-sample R^2 % (CW p)	Slope, 12 months (t)	IVX p, 12 months	Next to the forward premium: slope (IVX p)	AR(1) / δ
Carry*	0.15	0.20	2.70	4.79	.041	1.03	0.31	3.93	.051	5.22	0.90 / 0.06
	(0.14)		(1.33)	(1.98)			(.078)	(3.20)		(.072)	
FV: inflation 1y+3y	0.27	0.20	4.44	7.17	.012	1.52	0.90	5.07	.046	7.97	0.97 / 0.04
	(0.26)		(1.56)	(2.39)			(.037)	(3.15)		(.023)	
π^*: 3y inflation + r^*	0.18	0.20	3.05	4.79	.072	0.95	0.49	3.48	.136	5.17	0.97 / 0.01
	(0.17)		(1.32)	(1.70)			(.091)	(2.20)		(.142)	
π^*: trend inflation	0.31	0.18	5.21	5.64	.007	1.78	1.46	3.98	.034	6.34	0.98 / 0.04
	(0.31)		(2.07)	(2.41)			(.018)	(3.00)		(.017)	
OLS: future yield	0.26	0.20	4.39	8.52	.010	1.60	1.00	6.04	.040	9.33	0.97 / 0.03
	(0.26)		(1.54)	(2.43)			(.038)	(3.28)		(.019)	
Hamilton	0.36	0.37	5.72	2.40	.094	0.56	0.32	2.72	.069	3.18	0.90 / 0.20
	(0.35)		(2.54)	(1.67)			(.072)	(3.03)		(.079)	
10-year yield	0.44	0.39	6.86	3.43	.024	1.09	0.84	3.00	.053	5.11	0.98 / 0.17
	(0.43)		(2.83)	(2.40)			(.009)	(3.02)		(.008)	
Carry	0.13	0.47	1.99	0.31	.392	0.06	−0.14	0.85	.109		0.55 / 0.05
	(0.10)		(0.71)	(0.57)			(.581)	(2.42)			

I. Macroeconomic drivers of the long-run rate

We use macroeconomic and demographic fundamentals from several sources. Following Ferreira and Shousha (2023), we measure productivity growth as the growth of the trend component of an average of total factor productivity, labor productivity and GDP per capita, drawing on the Long-Term Productivity Database of Bergeaud et al. (2016). The birth rate and the working-age share are taken from the United Nations World Population Prospects (2024 revision). The birth rate is the number of live births per 1,000 people per year, and the working-age share is the ratio of individuals aged 20 to 60 to the total population.

To show that the estimated long-run rates are connected to these secular forces, we estimate for each version of the long-run rate the annual panel regression

$$\bar{i}^{*}_{c,y} = a_c + b_1\, BR_{c,y} + b_2\, WA_{c,y} + b_3\, g_{c,y} + \varepsilon_{c,y}, \tag{6}$$

where $\bar{i}^{*}_{c,y}$ is the annual average of the long-run rate of country c in year y, $BR_{c,y}$ is the birth rate, $WA_{c,y}$ the working-age share and $g_{c,y}$ productivity growth. The country effects, a_c, mean that the slopes come from changes within each country over time. We report Driscoll-Kraay t-statistics, which allow for correlation across countries and over time.

Table I1 shows that the estimated long-run rates are related to the variables that the macroeconomic literature emphasizes as the main drivers of long-run rates (Carvalho et al., 2016; Cesa-Bianchi et al., 2022; Ferreira and Shousha, 2023; Obstfeld, 2023). For $carry^{*}$, the three variables explain 59% of the variation in the long-run rate within countries, against 64% for the 10-year yield itself and between 42% and 65% across the other versions.

The birth rate enters with a positive and significant slope in every version, with t-statistics between 4.53 and 9.66. For $carry^{*}$, one more birth per 1,000 people per year goes with a long-run rate 0.86 percentage points higher. A lower birth rate implies lower population growth, which raises the capital-labor ratio and depresses the marginal product of capital, much like a permanent slowdown in productivity growth, and so pushes down long-run interest rates (Carvalho et al., 2016).

Productivity growth also enters with a positive and significant slope in every version, with t-statistics between 3.35 and 4.77. There are at least two economic explanations for this link (Lunsford and West, 2019; Ferreira and Shousha, 2023). First, slower productivity growth reduces investment opportunities and therefore investment demand. Second, it lowers the expected income of households, which raises their precautionary savings. Within a savings and investment framework, both channels push long-run interest rates down.

Since the fair value uses GDP growth, part of this link is mechanical for $carry^*$. The link is as strong, however, for the two π^* versions, which use inflation only.

The working-age share, in contrast, is not significant at the 5% level once the birth rate and productivity growth are included, except for the OLS fit to the future yield, and its sign differs across versions. This is consistent with two channels that work in opposite directions. A falling working-age share acts like slower population growth and lowers long-run rates, but it also means more retirees, which lowers aggregate savings and raises real interest rates (Carvalho et al., 2016).

Table I1: Macro drivers of the long-run rate. Annual mean of each version's i^* on the birth rate, the working-age share and productivity growth, with country effects. Driscoll-Kraay t-statistics in parentheses. Versions as in Table A1.

Version	Birth rate	Working-age share	Productivity growth	R^2
Carry*	0.86	0.18	0.02	0.59
	(9.14)	(1.83)	(3.35)	
FV: inflation 1y+3y	0.80	0.14	0.02	0.58
	(7.66)	(1.47)	(4.08)	
π^*: 3y inflation + r^*	0.76	−0.24	0.03	0.46
	(5.07)	(−1.93)	(4.09)	
π^*: trend inflation	0.76	−0.18	0.04	0.42
	(4.53)	(−1.62)	(4.77)	
OLS: future yield	0.66	0.23	0.03	0.59
	(7.63)	(3.19)	(4.49)	
Hamilton	1.68	0.21	0.09	0.65
	(9.66)	(1.61)	(4.08)	
10-year yield	1.74	0.13	0.04	0.64
	(8.96)	(0.75)	(3.95)	